

Introduction to Number Theory

Koopa Koo

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Chapter 1

Divisibility

Number Theory studies the properties of *natural numbers* $1, 2, \dots$, also called the *positive integers*. The set of natural numbers, together with the set of negative integers and zero, form the set of *integers*. One unique feature of number theory is that it is very easy to state a problem in which the solution to the problem is immensely difficult. A famous example of this is known as the Fermat's Last Theorem, which says $x^n + y^n = z^n$ has no integer solutions other than $\pm 1, 0$ for $n \geq 3$. This problem has defeated the greatest minds in the history for more than 350 years, and it's solved in 1994 by Andrew Wiles. The proof is about 200 pages long, and was published in Annals of Mathematics. Equations in which we sought only integer solutions are called *Diophantine Equations*, which is of central interest in number theory. In this chapter, we will explore the world in number theory and answer a lot of interesting problems concerning the properties of integers.

1.1 Euclidean Algorithm

Definition 1.1. If $b = aq$ for some $q \in \mathbb{Z}$, then we say a divides b , and we write this as $a \mid b$.

Proposition 1.2 (Properties of Divisibility). Suppose $a, b \in \mathbb{Z}$. Then:

(a) $a \mid b, b \mid c \Rightarrow a \mid c$.

(b) $d \mid a, d \mid b \Rightarrow d \mid ax + by$ for any $x, y \in \mathbb{Z}$, in particular $d \mid a + b, d \mid a - b$.

(c) If any two terms in $a + b = c$ are divisible by d , the third will also be divisible by d .

Proof. We will prove (a) and leave the rest as exercises. Suppose $a \mid b$, and $b \mid c$, then $b = aq_1, c = bq_2$ for some $q_1, q_2 \in \mathbb{Z}$. Hence $c = aq_1q_2$, hence $a \mid c$ for $q_1q_2 \in \mathbb{Z}$. $\square$

Proposition 1.3 (Division Algorithm). *For any integer a , when a is divided by b , then $a = qb + r$ with $0 \leq r < |b|$. The representation is unique.*

Definition 1.4. *Let $a, b \in \mathbb{Z}$ not both 0. The greatest common divisor (gcd) is the largest number d such that $d \mid a$ and $d \mid b$, and will be denoted $\gcd(a, b) = d$ or just $(a, b) = d$. The least common multiple (lcm) is the smallest number ℓ such that $a \mid \ell$ and $b \mid \ell$, and will be denoted $\text{lcm}(a, b) = \ell$ or just $[a, b] = \ell$.*

Definition 1.5. *For $a, b \in \mathbb{Z}$, they are called relatively prime if $(a, b) = 1$.*

The following theorem is very important and useful, the proof of this theorem uses Euclidean Algorithm which we will demonstrate shortly.

Theorem 1.6. *For $a, b \in \mathbb{Z}$, then there exists $x, y \in \mathbb{Z}$ such that*

$$ax + by = d = (a, b).$$

In particular, if $(a, b) = 1$, then there exists $x, y \in \mathbb{Z}$ such that

$$ax + by = 1.$$

Euclidean Algorithm is basically repeated use of division algorithm and it is very handy in finding the gcd of two numbers.

Example 1.1. Find $(121, 7)$. To begin, write $121 = 7 \times 17 + 2$, then we have:

$$121 = 17 \times 7 + 2$$

$$7 = 3 \times 2 + 1$$

$$2 = 2 \times 1 + 0$$

$(121, 7)$ is the least non-zero remainder, which is 1, thus $(121, 7) = 1$.

Also, to find the x, y in the theorem. We run the algorithm backwards:

$$1 = 7 - 3 \times 2$$

$$1 = 7 - 3 \times (121 - 7 \times 17) = 7 - 3 \times 121 + 51 \times 7 = 52 \times 7 - 3 \times 121.$$

Note that We substitute in $2 = 121 - 7 \times 17$. Hence x, y are $-3, 52$ respectively.

Note also that such x, y we found are NOT unique. Can you find one more pair that works? Can you find infinitely many? It is a good exercise to verify that such x, y we found are NOT unique, and for this particular example, find infinitely such pairs that works.

As a consequence of this example, we have the following:

Proposition 1.7. *Suppose $a = qb + r$, then $(a, b) = (b, r)$.*

Proof. Let $d = (a, b)$. Then $d \mid a, d \mid b$ implies $d \mid r$. Hence $d \mid (b, r)$. Now let, $e = (b, r)$. Then $e \mid b, e \mid r$ implies $e \mid a$. Hence $e \mid d$. Therefore $d = e$. $\square$

Lemma 1.8. Suppose $a, m, n \in \mathbb{N}$. Then we have: $a^m - 1 \mid a^n - 1$ iff $m \mid n$.

Proof. Write $n = mq + r$. Then we have:

$$\frac{a^n - 1}{a^m - 1} = a^r \frac{a^{mq} - 1}{a^m - 1} + \frac{a^r - 1}{a^m - 1}.$$

Now, since $a^{mq} - 1 = (a^m - 1)((a^m)^{q-1} + (a^m)^{q-2} + \dots + 1)$. We have $a^m - 1 \mid a^n - 1$ iff $a^m - 1 \mid a^r - 1$. However, $r < m$ implies $r = 0$. Hence the claim. $\square$

Lemma 1.9. Suppose $a, m, n \in \mathbb{N}$. Then we have $(a^m - 1, a^n - 1) = a^{(m,n)} - 1$.

Proof. Suppose $(n, m) = 1$ and write $n = mq_1 + r_1, 0 < r_1 < m$. We have:

$$a^m - 1 = a^r(a^{mq} - 1) + (a^r - 1) = q'_1(a^m - 1) + (a^{r_1} - 1).$$

By proposition 1.7, we have $(a^n - 1, a^m - 1) = (a^m - 1, a^{r_1} - 1)$. Next write $m = q_2 r_1 + r_2, 0 < r_2 < r_1$. Then following the exact same argument we have: $(a^m - 1, a^{r_1} - 1) = (a^{r_1} - 1, a^{r_2} - 1)$. Continuing in this manner we would have $(a^m - 1, a^n - 1) = (a^{r_n} - 1, a^{r_{n+1}} - 1)$. Since $(m, n) = 1, r_n = 1$ for sufficiently large n . Hence $(a^m - 1, a^n - 1) = (a^{r_n} - 1, a - 1) = a - 1$, which proves the special case where $(m, n) = 1$. Next suppose $(m, n) = d$. Since $(m/d, n/d) = 1$, we have $(a^m - 1, a^n - 1) = ((a^d)^{m/d} - 1, ((a^d)^{n/d} - 1)) = a^d - 1$ as claimed. $\square$

The next example will show you that consecutive Fibonacci Numbers are relatively prime. Recall that Fibonacci Numbers are defined by $f_1 = f_2 = 1, f_{n+2} = f_{n+1} + f_n$ for $n \geq 1$.

Example 1.2. Prove that $(f_n, f_{n+1}) = 1$ for all n .

At a first glance, one might be clueless on how one should approach the problem. We could try small cases and convince ourselves that the statement is correct. However, we cannot exhaust all cases, and we do not have a proof. Well, how about applying the Euclidean Algorithm and see what we get. But wait, how can we apply Euclidean Algorithm when we do not know the numbers f_n and f_{n+1} ? Well, what else do we know about f_n and f_{n+1} ? Are there any relations between the two numbers? Sure there are, we have: $f_{n+1} = f_n + f_{n-1}$, i.e. this is: $f_{n+1} = 1 \times f_n + f_{n-1}$ and now we use Proposition 1.7 to get $(f_{n+1}, f_n) = (f_n, f_{n-1})$. Now keep reducing until, $n = 1$ giving $(f_{n+1}, f_n) = (f_2, f_1) = (1, 1) = 1$. We are done.

Example 1.3. Suppose $x + y = 104055, (x, y) = 6937$. Find all such $x, y \in \mathbb{N}$, and $x < y$.

Write $x = 6937a, y = 6937b$ where $a, b \in \mathbb{N}$ with $(a, b) = 1$. Note that $(a, b) = 1$ otherwise it would contradict the definition of 6937 being the GREATEST common divisor of x, y . Thus this gives: $a + b = 15$ Now, since $(a, b) = 1$ we are down to the following possibilities: $(a, b) = \{(1, 14), (2, 13), (4, 11), (7, 8)\}$. Plugging this back in gives the appropriate x, y .

Example 1.4 (IMO 1959 Q1). Prove that $(21n + 4, 14n + 3) = 1$ for all n .

Applying the Euclidean algorithm, we have:

$$\begin{aligned} 21n + 4 &= 1 \times (14n + 3) + (7n + 1) \\ 14n + 3 &= 2 \times (7n + 1) + 1 \\ 7n + 1 &= (7n + 1) \times 1 + 0 \end{aligned}$$

Thus $(21n + 4, 14n + 3) = 1$ for all n .

Alternatively, let $d = (21n + 4, 14n + 3)$, then we have: $21n + 4 = da$, $14n + 3 = db$. Note that $a > b$. Also, $2(21n + 4) - 3(14n + 3) = -1$ yields $d(3b - 2a) = 1$. This shows that $d = 1$ and $3b - 2a = 1$ for both of them are natural numbers. Hence we have $d = 1$ as desired.

Example 1.5. Find all $n \in \mathbb{N}$ such that $n + 1 \mid n^2 + 1$.

To see this, note that $n^2 - 1 = (n + 1)(n - 1)$ thus $n + 1 \mid n^2 - 1$. So, if $n + 1 \mid n^2 + 1$ and $n + 1 \mid n^2 - 1$, then $n + 1 \mid (n^2 + 1) - (n^2 - 1) = 2$. Thus we have $n + 1 = 2$ or $n = 1$ is the only such n .

Alternatively, in order for $n + 1 \mid n^2 + 1$ we need $(n^2 + 1)/(n + 1)$ to be an integer. Let $m = n + 1$, then $n = m - 1$, thus $n^2 + 1 = (m - 1)^2 + 1 = m^2 - 2m + 2$. Hence we have:

$$\frac{n^2 + 1}{n + 1} = \frac{m^2 - 2m + 2}{m} = m - 2 + \frac{2}{m}.$$

In order for this expression to be an integer, we need $2/m$ to be an integer. Thus $m = 1, 2$. But since $m = 1$ gives $n = 0$, which is not natural. Hence $m = 2$, which gives $n = 1$ is the only solution.

Example 1.6. Find $d = (n^2 + 3n + 2, 6n^3 + 15n^2 + 3n - 7)$.

Proof. Since $6n^3 + 15n^2 + 3n - 7 = 6n(n^2 + 3n + 2) + (-3n^2 - 9n - 7)$, we have $d = (n^2 + 3n + 2, 3n^2 + 9n + 7)$. Next, we have $3n^2 + 9n + 7 = 3(n^2 + 3n + 2) + 1$. Hence $d = (n^2 + 3n + 2, 1) = 1$. $\square$

Proposition 1.10. For $a, b \in \mathbb{N}$, we have

$$ab = (a, b)[a, b].$$

Theorem 1.11 (Legendre). Let $\nu_a(n)$ denotes the highest power of a dividing n . i.e. $\nu_p(n!) = e$ if $n! = p^e m$, where p is a prime and $(p, m) = 1$. Then we have:

$$\nu_p(n!) = \sum_{i=1}^{\infty} \left[\frac{n}{p^i} \right],$$

here $[x]$ denotes the greatest integer not exceeding x . i.e. $[2, 99] = 2$, $[2.01] = 2$.

Proof. If $p^i > n$, then $[n/p^i] = 0$. We proceed by induction on n . The case $n = 1$ is trivial. Now suppose that the result is true for $n = k$. We try $n = k + 1$. Let j be the highest power of p dividing $n + 1$. Since $(k + 1)! = (k + 1)k!$, it suffices to prove that $\sum [(n + 1)/p^i] - \sum [n/p^i] = j$. However, we have:

$$\left[\frac{n+1}{p^i} \right] - \left[\frac{n}{p^i} \right] = \begin{cases} 1 & \text{if } p^i \mid n+1 \\ 0 & \text{if } p^i \nmid n+1 \end{cases}$$

and hence

$$\sum \left[\frac{n+1}{p^i} \right] - \sum \left[\frac{n}{p^i} \right] = j.$$

The last equality holds because j is the number of times such that $p^i \mid n + 1$. This completes the induction and we are done. $\square$

Example 1.7. Find the highest power of 3 dividing $100!$.

By Legendre's theorem, $\nu_3(100) = [100/3] + [100/9] + [100/27] + [100/81] = 33 + 11 + 3 + 1 = 48$. Hence $100! = 3^{48}m$, where $(3, m) = 1$.

Example 1.8. Prove that p divides $\binom{p}{k}$ for $0 < k < p$.

Note that $\binom{p}{k} = p!/[k!(p-k)!]$. It suffices to show that $\nu_p(p!) - \nu_p(k!) - \nu_p((p-k)!) > 0$. Since p is a prime, we have $[k/p] = 0$ if $0 < k < p$. Thus $\nu_p(p!) - \nu_p(k!) - \nu_p((p-k)!) = 1 - 0 - 0 = 1 > 0$. Done.

1.2 Problems

Problem 1.1. Prove that $a - c \mid ab + cd \Rightarrow a - c \mid ad + bc$.

Problem 1.2. Find $(2n + 3, n + 7)$. [Hint: your answer should depend on whether 11 divides $n - 4$.]

Problem 1.3. Let $a, b \in \mathbb{Z}$ with $(a, b) = 1$. Prove that: (a) $(a + b, a - b) = 1$ or 2 .

(b) $(2a + b, a + 2b) = 1$ or 3 .

(c) $(a^2 + b^2, a + b) = 1$ or 2 .

(d) $(a + b, a^2 - 3ab + b^2) = 1$ or 5 .

Problem 1.4. Let $a, b \in \mathbb{Z}$ with $(a, b) = 1$. Find possible values of:

(a) $(a^3 + b^3, a^3 - b^3)$

(b) $(a^2 - b^2, a^3 - b^3)$.

Problem 1.5. Suppose $b \mid a$, show that $2^b - 1 \mid 2^a - 1$.

Problem 1.6. Show that not both integers: $2^n - 1, 2^n + 1$ can be prime.

Problem 1.7 (Putnam 2000). Prove that the expression

$$\frac{\gcd(m, n)}{n} \binom{n}{m}$$

is an integer for all pairs of integers $n \geq m \geq 1$.

Problem 1.8. Suppose a, b are positive integers and $b^2 + ab + 1$ divides $a^2 + ab + 1$. Prove that $a = b$.

Problem 1.9. Show that $n!$ is divisible by $1 + 2 + \dots + n$ iff $n + 1$ is not an odd prime.

Problem 1.10. Prove that $n!(n - 1)! \mid (2n - 2)!$ for all $n \in \mathbb{N}$.

Problem 1.11. Suppose $a, b \in \mathbb{N}$. Prove that $a!b!(a + b)! \mid (2a)!(2b)!$.

Problem 1.12. Given $a, b, n \in \mathbb{N}$ with $n \mid a^n - b^n$. Prove that $n \mid (a^n - b^n)/(a - b)$.

Chapter 2

Primes

A positive integer is a *prime* number if it is only divisible by itself and 1. An integer is *composite* if it is a product of two integers each with absolute value greater than 1. Primes are the building blocks of integers because we can uniquely factorize each integer into its prime factors up to reordering the factors and multiplication by ± 1 . We begin with the following theorem which is taught in elementary school:

2.1 Fundamental Theorem of Arithmetic

Theorem 2.1 (Fundamental Theorem of Arithmetic). *Suppose $n \in \mathbb{N}$, then $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$, where p_i are distinct prime factors of n . The representation is unique up to reordering of the factors.*

As a side note, it is because of this theorem that we do not treat 1 as a prime because of uniqueness of factorization. If 1 were a prime, then we will lose the uniqueness because $6 = 1 \cdot 2 \cdot 3 = 1 \cdot 1 \cdot 2 \cdot 3$. Hence by convention, the first prime is 2. However, 1 has a special name as *unit*.

There is a question you may want to ask, that is, how many prime numbers are there? The answer is not very surprising:

Theorem 2.2 (Infinitude of primes). *There are infinitely many prime numbers.*

The proof is originally due to Euclid, and it is perhaps one of the earliest mathematical proofs.

Proof. We will prove this by contradiction. That is suppose there are only finitely many primes, list them as: $\{p_1, p_2, \dots, p_k\}$. Now we consider the number: $N = p_1 p_2 \dots p_k + 1$. Then it is obvious that p_i does not divide N for if they do, then they also divide 1, which is absurd. However, since N is not divisible by any prime in our list, which consists of all

primes by assumption. This implies N must itself be a prime. However, N is certainly not in our list for it is bigger than any p_i in our list. This is a contradiction that our list is complete. Hence the assumption that there are only finitely many primes is false, and thus there are infinitely many primes. $\square$

The next proposition is an exercise for practicing this kind of argument.

Proposition 2.3. *There are infinitely many primes p of the form $p = 4k + 3$ for some k .*

Proof. Leave as an exercise. [Hint: Let $N = 4p_1p_2 \dots p_k + 3$ and mimic Euclid's proof.] $\square$

Theorem 2.4 (Euclid's Lemma). *Suppose p is a prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. Suppose $p \nmid a$, then I claim that $p \mid b$. Since $p \nmid a$ and p is prime, we have $(p, a) = 1$ thus there exists $x, y \in \mathbb{Z}$ such that $px + ay = 1$. Multiplying both sides by b gives: $bpx + aby = b$, now $p \mid bpx, p \mid aby \Rightarrow p \mid b$. This finishes the proof. $\square$

Theorem 2.5 (Euler's Product). *Let $s \in \mathbb{C}$, with $\operatorname{Re} s > 1$ then*

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ primes}} \frac{1}{1 - p^{-s}},$$

where p runs over all prime numbers.

Remark: The function $\zeta(s)$ defined in theorem 2.5 is the famous **Riemann Zeta Function** related to the Riemann Hypothesis, one of the 7 millennium problems.

To test whether a number is prime, one naive way is to test whether n is divisible by any of the prime number preceding n . In fact, one can do a lot better, as the next example illustrates:

Example 2.1. If n is a composite integer, then n has a prime factor not exceeding $\sqrt{n}$.

Proof. We proceed by contradiction, suppose $n = ab$, where both $a, b > 1$. We either have $a \leq \sqrt{n}$, or $n \leq \sqrt{n}$. Otherwise both $a, b > \sqrt{n}$ implies $ab > n$. WOLOG (without loss of generality), assume $a \leq \sqrt{n}$. If a is prime, then we are finished, otherwise a has a prime factor $p < a \leq \sqrt{n}$. $\square$

This example shows that in order to test whether a number n is prime, we only need to verify all the prime divisors of n up to $\sqrt{n}$. For instance, to see whether 2007 is prime, we only need to verify whether any prime $p < \sqrt{2007} < 45$ divides 2007.

Proposition 2.6. 1. *Suppose $a \mid bc$ and $(a, b) = 1$ Prove that $a \mid c$.*

2. *Suppose $ab = n^k$ and $(a, b) = 1$ Prove that a, b are itself k -th powers.*

Example 2.2. Show that $m = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ is not an integer for $n > 1$.

Proof. This is a tough one, the idea here is to look at powers of 2. We again proceed by contradiction. First assume m is an integer. Let k be the highest power of 2 dividing n . Thus we have $2^k \leq n < 2^{k+1}$. Let t be the least common multiple of $1, 2, \dots, n$ excluding 2^k . Look at $tm = t + \frac{t}{2} + \cdots + \frac{t}{n}$, notice that every term on the right hand side is an integer except $t/2^k$. Hence the right hand side is not an integer, but m is an integer by assumption and thus tm must be an integer. This is a contradiction and we are done. $\square$

We end this section by a theorem due to Erdos:

Theorem 2.7 (Four Number Theorem). *Suppose a, b, m, n are positive integers such that:*

$$an = bm,$$

then there exists positive integers s, t, u, r such that

$$\begin{aligned} a &= rs, & n &= tu, \\ b &= rt, & m &= su. \end{aligned} \tag{2.1}$$

2.2 Problems

Problem 2.1. Prove Proposition 2.3. Explain why the same proof does not work for primes of the form $4k + 1$.

Problem 2.2. Prove Proposition 2.6.

Problem 2.3. Suppose $p, p^2 + 2$ are primes. Show that $p^3 + 2$ is also prime. [Hint: note that p is either 3, $3k + 1$, or $3k - 1$ for some k .]

Problem 2.4. Show that $p^2 + q^3 = r^4$ has no solutions for primes p, q, r .

Problem 2.5. Suppose $2^p - 1$ is prime. Prove that p is prime. [Hint: Argue by contradiction, i.e. suppose p is composite, (i.e. $p = ab, a, b > 1$), and show that $2^p - 1 = AB, A, B > 1$.]

Problem 2.6. Show that $m = 1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n+1}$ is not an integer for $n > 1$.
[Hint: look at powers of 3.]

Problem 2.7. Assuming convergence, prove theorem 2.5 using Fundamental theorem of Arithmetic.

The following exercises give alternative proofs that there are infinitely many primes.

Problem 2.8. a) Show that

$$\log \zeta(s) = \sum_{p \text{ prime}} p^{-s} + \sum_{p \text{ prime}} \sum_{n=2}^{\infty} \frac{p^{-ns}}{n}.$$

b) Show that

$$\sum_{p \text{ prime}} \sum_{n=2}^{\infty} \frac{p^{-ns}}{n} < 1.$$

c) By letting $s \rightarrow 1$, and the fact that the harmonic series diverges. Show that

$$\sum_{p \text{ prime}} \frac{1}{p}$$

diverges.

Problem 2.9. Prove that $2^{2^n} - 1$ has at least n distinct prime factors. Hence deduce that there are infinitely many prime numbers.

Chapter 3

Congruence

The notion of congruence first appeared in Gauss's celebrated book: "*Disquisitiones Arithmeticae*." It is a remarkable invention of Gauss for it tremendously simplifies many problems concerning divisibility. To exemplify the importance of congruence, it is reasonable to say that in order to master number theory, getting used to the notion of congruence is the first step.

3.1 Modular Arithmetic

Definition 3.1. $a \equiv b \pmod{m}$ iff $m \mid a - b$, in which we say that a is congruent to b modulo m .

Example 3.1. In congruence notation, we have: $3 \equiv 1 \pmod{2}$, $7 \equiv 4 \equiv 1 \equiv -2 \equiv 10 \pmod{3}$. Notice that n is even iff $n \equiv 0 \pmod{2}$.

Proposition 3.2 (Basic Properties of Congruence). Suppose $a, b, c, d, m \in \mathbb{Z}$, we have:

1. $a \equiv b \pmod{m}, b \equiv c \pmod{m} \Rightarrow a \equiv c \pmod{m}$
2. $a \equiv b \pmod{m}, c \equiv d \pmod{m} \Rightarrow a \pm c \equiv b \pm d \pmod{m}$.
3. $a \equiv b \pmod{m}, c \equiv d \pmod{m} \Rightarrow ac \equiv bd \pmod{m}$.
4. $ab \equiv ac \pmod{m}, (a, m) = 1 \Rightarrow b \equiv c \pmod{m}$.

Proof. Leave as an exercise. □

Remark: Note that the condition $(a, m) = 1$ in the cancelation law (i.e. part 4) is absolutely crucial. For example, if you take $a = 4, m = 2, b = 1, c = 2$ then the result is false, $b \not\equiv c \pmod{2}$.

Basically, besides division, the arithmetic of congruences is exactly the same as the ordinary arithmetic.

As a quick application of congruences, we have the following facts, which you learned from elementary school. (But I guess you did not know why they were true then.)

Notation 1. We let $N = \overline{a_n a_{n-1} \dots a_0}$, where $a_i \in 0, 1, 2, \dots, 9$ to denote a $n + 1$ digit number.

Example 3.2. Notice that $N = \overline{a_n a_{n-1} \dots a_0} = a_n \times 10^n + \dots 10 \times a_1 + a_0$.

Proposition 3.3 (Divisibility Tests). *Let $N = \overline{a_n a_{n-1} \dots a_0}$.*

1. $N \equiv 0 \pmod{3}$ iff $a_n + a_{n-1} + \dots + a_0 \equiv 0 \pmod{3}$.
2. $N \equiv 0 \pmod{9}$ iff $a_n + a_{n-1} + \dots + a_0 \equiv 0 \pmod{9}$.
3. $N \equiv 0 \pmod{11}$ iff $a_0 - a_1 + a_2 - \dots + (-1)^n a_n \equiv 0 \pmod{11}$.
4. $N \equiv 0 \pmod{4}$ iff $4 \mid \overline{a_1 a_0}$.
5. $N \equiv 0 \pmod{8}$ iff $8 \mid \overline{a_2 a_1 a_0}$.

To restate in words, part 1 says: N is divisible by 3 if and only if the sum of the digits of N is divisible by 3. In other words, to determine whether a number is divisible by 3, it suffices to check whether the sum of the digits is divisible by 3.

Proof. Recall that $N = a_n \times 10^n + \dots + a_0$, also since $10 \equiv 1 \pmod{3}$, we have $10^n \equiv 1^n \equiv 1 \pmod{3}$. Hence we have: $N \equiv a_n \times 10^n + a_{n-1} \times 10^{n-1} + \dots + 10a_1 + a_0 \equiv a_n + a_{n-1} + \dots + a_1 + a_0 \pmod{3}$.

For the other proofs, We will leave them as exercises. Hint: $10^2 \equiv 0 \pmod{4}$, and $10^3 \equiv 0 \pmod{8}$. $\square$

Remark: Not only does Proposition 3.3 provides a way to determine whether a number is divisible by 3, 9, 11, 4, and 8 respectively. It says more, it actually determines the remainder upon division by 3, 9, 11, 4 and 8 respectively.

Example 3.3. Determine the remainder when $N = 1234567891$ is divisible by 3.

By Proposition 3.3, it suffices to determine the sum of digits mod 3. We thus have: $N \equiv 1 + 2 + \dots + 9 + 1 \equiv 46 \equiv 4 + 6 \equiv 10 \equiv 1 + 0 \equiv 1 \pmod{3}$. Hence the remainder is 1.

Note that in the previous example, we used Proposition 3.3 three times.

Example 3.4 (IMO). Let A be the sum of digits of 4444^{4444} , B be the sum of digits of A , and C be the sum of digits of B . Find C .

So, how should we do this problem? As before, we know that $4444^{4444} \equiv A \equiv B \equiv C \pmod{9}$. We also notice that $4444^{4444} > A > B > C$. So how can we determine C ? Well, if we know that $C < 10$ then we can determine it. To this end, we need to limit the size of A, B and C . So how big can A be? Well, $4444^{4444} < 10000^{4444}$ which is a $4 \times 4444 + 1 = 17777$ digits number. Hence 4444^{4444} is at most 17777 digits. In which case, $A \leq 9 \times 17777 = 159993$. Thus $B \leq 1 + 9 + 9 + 9 + 9 + 9 = 46$. Hence $C < 4 + 9 = 13$. However, we have: $4444 \equiv 4 + 4 + 4 + 4 \equiv 16 \equiv 7 \pmod{9}$. Thus, $4444^{4444} \equiv 7^{4444} \pmod{9}$. Now we have $7^3 \equiv 343 \equiv 1 \pmod{9}$. Also, $4444 \equiv 16 \equiv 1 + 6 \equiv 1 \pmod{3}$. Write $4444 = 3k + 1$, we have $7^{4444} = 7^{3k+1} = 7^{3k} \times 7 = (7^3)^k \times 7 \equiv 1^k \times 7 \equiv 7 \pmod{9}$. Therefore, we have $C \equiv 7 \pmod{9}$ and $C < 13$. Finally we may conclude that $C = 7$.

Up to now, it seems like calculating higher power congruence plays a central role in some of the problems. (i.e. the previous IMO problem). Are there any theorems that make our lives easier? The answer is yes, and it is the topic of the next section.

3.2 Fermat's Little Theorem and Euler's Theorem

Theorem 3.4 (Fermat's Little Theorem). *Let p be a prime and $p \nmid a$, then we have:*

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. Look at $\{1, 2, \dots, p-1 \pmod{p}\}$. Suppose $p \nmid a$. Since $(a, p) = 1$, we have $ai \equiv aj \pmod{p}$ iff $i \equiv j \pmod{p}$. This says $\{a, 2a, \dots, a(p-1) \pmod{p}\}$ is equal to $\{1, 2, \dots, p-1 \pmod{p}\}$ as sets. Multiplying all the elements in both sets and equating gives:

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Finally, $(p, (p-1)!) = 1$ gives $a^{p-1} \equiv 1 \pmod{p}$. □

Definition 3.5 (Euler φ function). *Let $\varphi(m) = \{\text{number of integers, } a \text{ less than } m \text{ and relatively prime to } m\}$. i.e. $\varphi(m) = \{a \in \mathbb{N} \mid (a, m) = 1 \text{ and } a < m\}$.*

Example 3.5. $\varphi(4) = 2$, for 1, 3 are the integers less than 4 and relatively prime to 4. $\varphi(5) = 4$.

Theorem 3.6 (Formula for $\varphi(m)$). *Let $m = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$, where the p_i are distinct prime factors of m then:*

$$\varphi(m) = m \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right) = m \left(1 - \frac{1}{p_1}\right) \times \cdots \times \left(1 - \frac{1}{p_k}\right).$$

Proof. We Divide m on both sides and interpret the formula probabilistically. The left hand side is $\varphi(m)/m$ which is the probability of randomly selecting a number $1 \leq a \leq m$ such that $(a, m) = 1$. On the other hand, this is the same as selecting a number $1 \leq a \leq m$ such that a is not divisible by any of the prime divisor of m . Fix a prime divisor p of m , the probability that $1 \leq a \leq m$ is not divisible by p is:

$$\frac{m - m/p}{m},$$

where the denominator is the total number less than or equal to m that is not a multiple of p . Now multiply all these product gives the probability of selecting a number $1 \leq a \leq m$ such that a is not divisible by any prime p dividing m . Hence by calculating the same probability in two different ways, the assertion follows. $\square$

Example 3.6. Find $\varphi(100)$.

Since $100 = 2^2 5^2$, we have $\varphi(100) = 100(1 - 1/2)(1 - 1/5) = 100(1/2)(4/5) = 40$.

Definition 3.7 (Multiplicative Function). *A function f is called a multiplicative function if it satisfy the property:*

$$f(mn) = f(m)f(n),$$

whenever $(m, n) = 1$.

Theorem 3.8. $\varphi(m)$ is a multiplicative function.

Proof. This follows from straightforward calculation using the formula for $\varphi(m)$. $\square$

Theorem 3.9 (Euler's Theorem). *Suppose $(a, m) = 1$ then we have:*

$$a^{\varphi(m)} \equiv 1 \pmod{m}.$$

Proof. Let $S = \{r_1, r_2, \dots, r_{\varphi(m)} \pmod{m}\}$ be the set of integers less than m and relatively prime to m . Then $(a, m) = 1$ implies $ar_i \equiv ar_j \pmod{m}$ iff $r_i \equiv r_j \pmod{m}$. This says the set $\{ar_i, ar_2, \dots, ar_{\varphi(m)} \pmod{m}\}$ is the set S . Multiplying all elements in both sets and equating gives:

$$a^{\varphi(m)} \prod_{i=1}^{\varphi(m)} r_i \equiv \prod_{i=1}^{\varphi(m)} r_i \pmod{m}.$$

Since $(\prod r_i, m) = 1$ we may cancel the product from both sides and we get:

$$a^{\varphi(m)} \equiv 1 \pmod{m}.$$

□

As an application of Euler's theorem, we go back to the previous IMO problem.

Example 3.7. Find $4444^{4444} \pmod{9}$.

Since $4444 \equiv 7 \pmod{9}$, $(7, 9) = 1$, and $\varphi(9) = 9(1 - 1/3) = 9(2/3) = 6$, we have $7^6 \equiv 1 \pmod{9}$ by Euler's theorem. Now $4444 \equiv 4 \pmod{6}$. By writing $4444 = 6k + 4$, we have $4444^{4444} \equiv 7^{4444} \equiv 7^{6k+4} \equiv 7^4 \equiv 7 \pmod{9}$.

Example 3.8. Suppose p is a prime such that $p \equiv 2 \pmod{3}$. Prove that for every integer n , there exists integer a such that $n \equiv a^3 \pmod{p}$.

Proof. (Due to Carto Wong) The statement is obvious if $p|n$, hence assume $(p, n) = 1$. Since $p \equiv 2 \pmod{3}$, we have $(p-1, 3) = 1$. Hence there exists $x, y \in \mathbb{Z}$ such that $(p-1)x + 3y = 1$. By Fermat's little theorem, $n^{p-1} \equiv 1 \pmod{p}$. Therefore,

$$n \equiv n^1 \equiv n^{(p-1)x+3y} \equiv n^{(p-1)x} (n^y)^3 \equiv a^3 \pmod{p},$$

where $a = n^y$.

□

Example 3.9. Given p is a prime and suppose $a^p \equiv b^p \pmod{p}$. Prove that $a^p \equiv b^p \pmod{p^2}$.

Proof. First of all, if $p|a$, then the assertion obviously holds. Now suppose $(p, a) = 1$, then by Fermat's little theorem, we have $a \equiv b \pmod{p}$. Hence $a = pk + b$ for some k . This implies $a^p = (b + pk)^p = b^p + pb^{p-1}(pk) + p^2M$, for some M by binomial theorem. Thus $a^p \equiv b^p \pmod{p^2}$. □

Corollary 3.10 (Exponentiation Dream). *Let p be a prime, then $(a+b)^p \equiv a^p + b^p \pmod{p}$, for all $a, b \in \mathbb{Z}$.*

Proof. By Fermat's little theorem, we have $(a+b)^p \equiv a+b \pmod{p}$. Again, by Fermat's little theorem, we have $a, b \equiv a^p, b^p \pmod{p}$. Hence $(a+b)^p \equiv a+b \equiv a^p + b^p \pmod{p}$ as claimed. □

Example 3.10. Prove that for a prime p , $2^p + 3^p$ is never a perfect power.

Proof. For $p = 2$, $2^2 + 3^2 = 13$ is not a perfect power. Suppose p is odd, then it is easy to see that $2^p + 3^p \equiv 0 \pmod{5}$. Next, we will show that $2^p + 3^p \not\equiv 0 \pmod{25}$ unless $p = 5$. To see this, suppose $2^p + 3^p \equiv 0 \pmod{25}$. Then we have: $2^p \equiv (-3)^p \equiv (22)^p \equiv 2^p 11^p \pmod{25}$. Since $(2^p, 25) = 1$, we have $11^p \equiv 1 \pmod{25}$. This implies $p \mid \varphi(25) = 20$. This implies $p = 5$. However, $2^5 + 3^5 = 275 = 25 \times 11$ is not a perfect power. Therefore, unless $p = 5$, otherwise we have $5 \mid 2^p + 3^p$ but $25 \nmid 2^p + 3^p$. Therefore, $2^p + 3^p$ is never a perfect power for all primes p . $\square$

Example 3.11 (IMO 2005). Let $a_n = 2^n + 3^n + 6^n - 1$. Determine all positive integers m that are relatively prime to a_n for all n .

Proof. We claim that for any prime p . There exists $n \in \mathbb{N}$ such that $a_n \equiv 0 \pmod{p}$. For $p = 2$, we have $a_1 = 10 \equiv 0 \pmod{2}$. For $p = 3$, we have $a_2 = 48 \equiv 0 \pmod{3}$. For $p \geq 5$. We consider $a_{p-2} = 2^{p-2} + 3^{p-2} + 6^{p-2} - 1$. Since $(6, p) = 1$, $a_{p-2} \equiv 0 \pmod{p}$ iff $6a_{p-2} \equiv 0 \pmod{p}$. Now, we have $6a_{p-2} = 32^{p-1} + 23^{p-1} + 6^{p-1} - 6 \equiv 3 + 2 + 1 - 6 \equiv 0 \pmod{p}$, by Fermat's little theorem. This conclude the claim. Therefore $m = 1$ is the required answer. $\square$

3.3 Chinese Remainder Theorem

The original form of the theorem, contained in a third-century book by a Chinese mathematician Sun Tzu and later republished in a 1247 book by Qin Jiushao. The following problem is contained in the book written by Sun Tsu:

There are certain things whose number is unknown. Repeatedly divided by 3, the remainder is 2; by 5 the remainder is 3; and by 7 the remainder is 2. What will be the number?

Remark: There are two side stories about Chinese Remainder Theorem, one historical about Han Shin of the Han Dynasty, and one fictional about a puzzle in a fiction on Chinese swordsman.

In this section, we would like to solve systems of linear congruence equations. We begin by a few examples:

Example 3.12. Find the smallest positive integer x such that when x is divided by 3 and 5, the remainder is 2 and 4 respectively.

We first translate the question into system of congruence equations:

$$\begin{aligned} x &\equiv 2 \pmod{3} \\ x &\equiv 4 \pmod{5} \end{aligned}$$

From the second equation, write $x = 5k + 4$ for some k , and then substitute x in the first equation gives: $5k + 4 \equiv 2 \pmod{3}$. This says $k \equiv 2 \pmod{3}$, which means $k = 3t + 2$

for some t . Thus $x = 15t + 14$. Choose $t = 0$ to obtain $x = 14$, being the minimal positive solution.

Example 3.13. Find the smallest positive integer x such that when x is divided by 3 and 6 the remainder is 2 and 3 respectively.

Imitate what we did in the previous example gives the system:

$$\begin{aligned}x &\equiv 2 \pmod{3} \\x &\equiv 3 \pmod{6}\end{aligned}$$

From the second equation, write $x = 6k + 3$ and substitute into the first equation gives: $6k + 3 \equiv 2 \pmod{3}$. This says $0 \equiv 2 \pmod{3}$, which is absurd, and the system has no solution.

From this example, we see that a system could have no solution, can we somehow determine that efficiently? The answer is yes and it's the content of the following theorem:

Theorem 3.11. *The 2 by 2 system:*

$$\begin{aligned}x &\equiv a_1 \pmod{m_1} \\x &\equiv a_2 \pmod{m_2}\end{aligned}$$

is solvable mod m_1m_2 iff $(m_1, m_2) | a_1 - a_2$. In particular, if $(m_1, m_2) = 1$, then the system is always solvable.

Proof. Imitate what we did in the earlier examples to obtain $m_2k + a_2 \equiv a_1 \pmod{m_1}$. Note that the system is solvable iff this congruence is solvable for k . From basic theory of linear congruence equation, $ax \equiv b \pmod{m}$ is solvable iff $(a, m) | b$. [For a proof of this, this is essentially Euclidean algorithm] Applying this to our situation here says the system is solvable iff $(m_1, m_2) | a_1 - a_2$. $\square$

After establishing this theorem, we may now state and prove the Chinese Remainder theorem:

Theorem 3.12 (Chinese Remainder Theorem). *Suppose $m_1, \dots, m_k$ are pairwise relatively prime. Then the k by k system:*

$$\begin{aligned}x &\equiv a_1 \pmod{m_1} \\x &\equiv a_2 \pmod{m_2} \\&\vdots \\x &\equiv a_k \pmod{m_k}\end{aligned}$$

has a solution mod $M = m_1m_2 \dots m_k$.

Before giving the proof, let's try to examine the $k = 3$ case. Since m_1, m_2, m_3 are pairwise relatively prime, we have $(m_1, m_2) = 1$ in particular. Applying theorem 3.11 to the system:

$$\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \end{aligned}$$

gives a solution $x \equiv a' \pmod{m_1 m_2}$. Now since $(m_1 m_2, m_3) = 1$. Applying theorem 3.11 to the system:

$$\begin{aligned} x &\equiv a' \pmod{m_1 m_2} \\ x &\equiv a_3 \pmod{m_3} \end{aligned}$$

gives a solution $x \equiv a \pmod{m_1 m_2 m_3}$. So the case $k = 3$ is settled.

Now we go back to prove the general case:

Proof. Proceed by induction. The case $k = 1$ is trivial, and the case $k = 2$ is precisely theorem 3.11. Suppose Chinese Remainder Theorem is true for k , try $k + 1$. Applying induction hypothesis to the system:

$$\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \\ &\vdots \\ x &\equiv a_k \pmod{m_k} \end{aligned}$$

gives a solution $x \equiv a' \pmod{m_1 m_2 \dots m_k}$. Now applying $k = 2$ case to the system:

$$\begin{aligned} x &\equiv a' \pmod{m_1 m_2 \dots m_k} \\ x &\equiv a_{k+1} \pmod{m_{k+1}} \end{aligned}$$

gives a solution $x \equiv a \pmod{m_1 m_2 \dots m_{k+1}}$. Hence we are done by induction. □

3.4 Polynomial Congruence, when the Power of Polynomial and Congruence Meet

In this section, we will study the following polynomial $f(x) = (x - 1)(x - 2) \dots (x - (p - 1))$ for a fixed prime p .

We begin with the following theorem:

Theorem 3.13. $f(x) \equiv x^{p-1} - 1 \pmod{p}$, for all primes p .

Proof. Consider $g(x) = f(x) - (x^{p-1} - 1)$. Then $g(x)$ is a polynomial of degree at most $p-2$. However, $g(x)$ has roots $1, 2, \dots, p-1 \pmod{p}$ by Fermat's little theorem. i.e. $g(x)$ has $p-1$ roots. Hence $g(x) \equiv 0 \pmod{p}$. $\square$

Corollary 3.14. *For a prime p , we have $(p-1)! \equiv -1 \pmod{p}$.*

Proof. The constant term of $f(x)$ is $(p-1)!$ and the constant term of $x^{p-1} - 1$ is -1 . Hence the claim. $\square$

Corollary 3.15. *Prove that the numerator of $n = 1 + \frac{1}{2} + \dots + \frac{1}{p-1}$ is divisible by p for a prime p .*

Proof. $1, \frac{1}{2}, \dots, \frac{1}{p-1}$ are roots of $x^{p-1}f(1/x)$. If we write $f(x) = x^{p-1} + a_{p-2}x^{p-2} + \dots + a_1x + a_0$. Then n equals the coefficient of x^{p-2} of $-x^{p-1}f(1/x)$, which equals $-a_1$. Now the coefficient of x^{p-2} in $x^{p-1}[(1/x)^{p-1} - 1]$ is obviously 0. Hence the claim. $\square$

Corollary 3.16. *Prove that the numerator of $n = 1 + \frac{1}{2} + \dots + \frac{1}{p-1}$ is divisible by p^2 for a prime $p \geq 5$.*

Proof. Following the proof of the previous corollary, it suffices to show that $a_1 \equiv 0 \pmod{p^2}$. By Viète's theorem, we have $a_0 = (p-1)!$. Now $f(p) = (p-1)! = p^{p-1} + a_{p-2}p^{p-2} + \dots + a_2p^2 + a_1p + (p-1)!$.

This gives $0 = p^{p-1} + a_{p-2}p^{p-2} + \dots + a_2p^2 + a_1p$. This implies: $a_2p^2 + a_1p \equiv 0 \pmod{p^3}$. By theorem 3.13, we have $p \mid a_2$ and thus $a_1p \equiv 0 \pmod{p^3}$. This implies $p^2 \mid a_1$ and we are done. $\square$

Theorem 3.17 (Wilson's Theorem).

$$(p-1)! \equiv -1 \pmod{p}$$

if and only if p is a prime.

Proof. We will give a different proof here. We first assume $(p-1)! \equiv -1 \pmod{p}$, and will show p is a prime. It is easy to check the result for $p = 2, 3$. Suppose $p \geq 5$, and suppose for the sake of contradiction that p is composite and write $p = ab$ with $a, b > 1$. Since $(p-1)! \equiv -1 \pmod{p}$ and a divides p , we have $(p-1)! \equiv -1 \pmod{a}$. Then note that since $a < p-1$ we have $(a)! \equiv 0 \pmod{a}$ and thus $(p-1)! = (p-1)(p-2)\dots(a+1)a! \equiv 0 \pmod{a}$. This is a contradiction unless $-1 \equiv 0 \pmod{a}$, which is absurd. Hence p must be prime.

Now suppose p is prime and we will show $(p-1)! \equiv -1 \pmod{p}$. To this end, for each $a \in \{1, 2, \dots, p-1 \pmod{p}\}$, there is a unique $x \in \{1, 2, \dots, p-1 \pmod{p}\}$ such that $ax \equiv 1 \pmod{p}$. Also, $a \equiv x \pmod{p}$ if $a^2 \equiv 1 \pmod{p}$, which happens if and only if $a \equiv \pm 1 \pmod{p}$. Thus $(p-1)! \equiv (1)(2)\dots(p-2)(p-1) \equiv p-1 \pmod{p}$. The last congruence follows because each of $2, 3, \dots, p-2$ is paired off with its inverse. This concludes the proof. $\square$

Corollary 3.18. Suppose $p = 4k + 1$ is a prime number, then $p \mid n^2 + 1$ for some n .

Proof. Pair up k and $p - k$ to get $k(p - k) \equiv -k^2 \pmod{p}$. Multiplying all these products for $k = 1, 2, \dots, (p - 1)/2$ gives:

$$(p - 1)! \equiv (-1)^{\frac{p-1}{2}} \left(\frac{p-1}{2} \right)^2 \pmod{p}.$$

By Wilson's Theorem, we have:

$$(-1)^{\frac{p-1}{2}} \left(\frac{p-1}{2} \right)^2 \equiv -1 \pmod{p}.$$

Hence if $p = 4k + 1$, we have $[(2k)!]^2 + 1 \equiv 0 \pmod{p}$.

□

3.5 Problems

Problem 3.1. Let $a_0 = 7, a_n = 7^{a_{n-1}}$ for $n \geq 1$. Find the unit digit of a_3 . [Hint: To find the unit digit of a_3 is equivalent to find $a_3 \pmod{10}$. Do you know why?]

Problem 3.2. Let $f(x) = a_n x^n + \dots + a_0 \in \mathbb{Z}[x]$. Prove that if $b \equiv c \pmod{m}$, then $p(b) \equiv p(c) \pmod{m}$. [Hint: your proof should be extremely short if you use Proposition 3.2 wisely.]

Problem 3.3. Prove that the numerator of $n = 1 + \frac{1}{2} + \dots + \frac{1}{p-1}$ is divisible by p for a prime p .

Problem 3.4. .

- a) Prove that $3 \mid 2^{2n+1} + 1$ for all $n \in \mathbb{N}$.
- b) Prove that $7 \mid 4^{3n+1} + 2^{3n+1} + 1$ for all $n \in \mathbb{N}$.

Problem 3.5. Find all $n \in \mathbb{N}$ such that $2^n - 1$ is divisible by 7, and prove that $2^n + 1$ is never divisible by 7.

Problem 3.6. Fix a prime p , and an integer a not divisible by p . Let k be the smallest integer such that $a^k \equiv 1 \pmod{p}$. Show that k divides $p - 1$.

Problem 3.7. Prove that n does not divide $2^n - 1$ for $n > 1$.

Problem 3.8. Let p, q be distinct primes. Suppose $a \not\equiv 1 \pmod{q}$ and $a^p \equiv 1 \pmod{q}$. Prove that $p \mid q - 1$.

Problem 3.9. Let $p > 3$ be a prime. Prove that $x^{p-1} + x^{p-2} + \cdots + x + 2$ is not a perfect square for any integer x .

Problem 3.10. The number: $21982145917308330487013369 = a^{13}$ for some positive integer a . Which one? (NO CALCULATORS!)

Problem 3.11. Given that $abcdef$ is a six-digit number (each letter represents one digit) and all of which are distinct and none of them equal to 0. Furthermore $abcdef$ satisfies the following equations:

$$bcdefa = 3 \times abcdef$$

$$cdefab = 2 \times abcdef$$

$$defabc = 6 \times abcdef$$

$$efabcd = 4 \times abcdef$$

$$fabcde = 5 \times abcdef$$

Find this unique number $abcdef$.

Problem 3.12. Define $a_n = 6^n + 8^n$. Find the remainder when a_{83} is divided by 49. [Hint: Either use Euler, or binomial theorem.]

Problem 3.13. Given any n show that there exists n consecutive composite numbers.

Problem 3.14. Suppose p, q are primes such that $q \nmid p - 1$. Prove there for all integer n , there exists $a \in \mathbb{Z}$ such that $n \equiv a^q \pmod{p}$.

Problem 3.15. If the 5-digit decimal number $a679b$ is a multiple of 72 find a and b .

Problem 3.16. Let $p \equiv 3 \pmod{4}$ be a prime. Prove that $a^2 + b^2 \equiv 0 \pmod{p}$ implies $a \equiv b \equiv 0 \pmod{p}$.

Problem 3.17. Let p be a prime other than 2 and 5. Let $a_k = 111 \dots 1$ be the number consists of k 1's. Prove that there are infinitely many k such that $a_k \equiv 0 \pmod{p}$.

Problem 3.18. Prove $24 \mid p^2 - 1$ for all primes $p \geq 5$.

Problem 3.19. Solve the following systems of congruence equations:

(a)

$$x \equiv 1 \pmod{7}$$

$$x \equiv 4 \pmod{9}$$

$$x \equiv 3 \pmod{5}$$

(b)

$$x \equiv 1 \pmod{6}$$

$$x \equiv 4 \pmod{9}$$

$$x \equiv 7 \pmod{15}$$

Problem 3.20. Let $n \in \mathbb{N}$ be given. Prove that there exists n consecutive positive integers such that each of them is not a prime power.

Problem 3.21. Explain why the proof of Corollary 3.16 fails when $p = 3$.

Problem 3.22. Following the notation in section 3.4, let $p \geq 5$ be a prime. Prove that $a_1 \equiv pa_2 \pmod{p^3}$.

Problem 3.23. Let p be an odd prime. Suppose $a_0, a_1, \dots, a_{p-1}$ is a complete residue system mod p , (i.e. it is a permutation of $0, 1, 2, \dots, p-1$.) suppose also that $b_0, b_1, \dots, b_{p-1}$ is also a complete residue system mod p . Prove that $a_0b_0, a_1b_1, \dots, a_{p-1}b_{p-1}$ cannot be a complete residue system mod p .

Chapter 4

Diophantine Equations

4.1 Linear Diophantine Equations

By linear Diophantine equations, we mean equations of the form:

$$ax + by = d,$$

where $a, b, d \in \mathbb{Z}$.

This is the easiest kind of Diophantine equation, and we can find all the solutions completely. We will illustrate the technique by a couple of examples and leave the reader to generalize the idea:

Example 4.1. Solve $2x + 3y = 1$ for integers.

Solution. Rewrite the equation as $x = (1 - 3y)/2 = (1 - y)/2 - y$. Since x and y are integers. This says $(1 - y)/2$ must be an integer and thus 2 divides $1 - y$. Write $1 - y = 2k$ for some integer k , then $y = 2k - 1$, and $x = k - (2k - 1) = 1 - k$. As k ranges over the integers, this gives all integer solutions to the equation. $\square$

Example 4.2. Solve $11x + 22y = 13$.

Proof. Imitate what did in the previous example, rewrite the equation as $x = (13 - 22y)/11 = 2/11 + 1 - 2y$. In order for x to be an integer, we need $2/11$ to be an integer, which is absurd. Hence the equation has no solution in integers. $\square$

Upon seeing the previous example, one might ask: Is there a more direct way of seeing that the equation is not solvable? The answer is yes, and it is the statement of the following theorem:

Theorem 4.1. *The equation $ax + by = d$ is solvable in integers if and only if $(a, b) \mid d$.*

Proof. Suppose $(a, b) \mid d$ then the existence of solution is asserted by a corollary of Euclidean Algorithm. Now suppose $ax + by = d$ is solvable. Let $D = (a, b)$. Then $D \mid a$ and $D \mid b$ implies $D \mid ax + by = d$. Done. $\square$

4.2 Is this a Square?

We will discuss some properties of perfect squares.

Theorem 4.2. *Let a^2 be a perfect square, then the following are true:*

1. $a^2 \equiv 0, 1, 4, 5, 6, 9 \pmod{10}$.
2. $a^2 \equiv 0, 1 \pmod{4}$.
3. $a^2 \equiv 0, 1, 4 \pmod{8}$.
4. $a^2 \equiv 0, 1 \pmod{3}$.
5. *There is no perfect square between two consecutive perfect square.*
6. *Non-zero number is a perfect square iff it has an odd number of factors.*
7. *Let p be a prime, if p divides a^2 , then p divides a .*

Proof. Leave as exercise. $\square$

Example 4.3. Let n be a positive integer and $n^4 + n^3 + n^2 + n + 1$ is a perfect square. Find all such n .

Proof. Let $k(n) = n^4 + n^3 + n^2 + n + 1$. If $k(n)$ is a square, so is $4k(n)$. Now $4k(n) = 4n^4 + 4n^3 + 4n^2 + 4n + 4 = n^2(4n^2 + 4n + 1) + 3n^2 + 4n + 4 = [n(2n + 1) + 1]^2 - n^2 + 2n + 3$. For $n \geq 3$, we have $-n^2 + 2n + 3 = 2 - (n - 1)^2 < 0$. Hence for $n \geq 3$, we have $[n(2n + 1)]^2 < 4k(n) < [n(2n + 1) + 1]^2$. Therefore, for $n \geq 3$, $4k$ is not a perfect square. Finally, we check that $k(1) = 5, k(2) = 31$. We conclude that $k(n)$ is never a square for positive integer n . $\square$

Example 4.4. Find all n such that $S = 2^8 + 2^{11} + 2^n$ is a perfect square.

Proof. First notice that it is convenient to factor out high powers of 2. To this end, we consider two cases: (1) $n \geq 8$ and (2) $n < 8$. If $n \geq 8$, then write $n = 8 + d$. Hence $S = 2^8(1 + 2^3 + 2^d)$. Since 2^8 is a square. It suffices to find d such that $1 + 2^3 + 2^d = 9 + 2^d$ is a square. Write $9 + 2^d = k^2$, then we have $2^d = k^2 - 3^2 = (k - 3)(k + 3)$. This implies $k - 3 = 2^a, k + 3 = 2^b$. Subtracting gives $6 = 2^b - 2^a$. Dividing 2 on both sides give $3 = 2^{b-1} - 2^{a-1}$. Therefore $b - 1 = 2, a - 1 = 0$. Thus $d = a + b = 4$. Now if $n < 8$. Considering mod 3 gives $S \equiv 2^n \equiv (-1)^n \pmod{3}$. For S to be a square, n must be even. Now check $n = 2, 4, 6$ respectively, we see that none of them is a square. Hence $n = 12$ is the only solution. $\square$

Example 4.5. Suppose n is a positive integer and d is a factor of $2n^2$. Prove that $n^2 + d$ is not a perfect square.

Proof. Let $2n^2 = kd$, for some k . If $n^2 + d = a^2$. Then we have $k^2a^2 = k^2(n^2 + d) = k^2n^2 + k^2d = n^2(k^2 + 2k)$. This implies $k^2 + 2k$ is a square. However, $k^2 < k^2 + 2k < (k+1)^2$. This implies $k^2 + 2k$ is not a square. Therefore $n^2 + d$ is not a square. $\square$

Example 4.6. Show that the sequence $11, 111, 1111, \dots$, contains no perfect square.

Proof. Every single term in the sequence is $3 \pmod{4}$. Hence none of them is a square. $\square$

Example 4.7. Given $a_n = 2 + 2\sqrt{28n^2 + 1}$ is a positive integer. Prove that a_n is a perfect square.

Proof. We have $4(28n^2 + 1) = (a_n - 2)^2 = a_n^2 - 4a_n + 4$. This implies $16 \times 7n^2 = a_n^2 - 4a_n$. From this we see that a_n is even. Write $a_n = 2b_n$. We then have: $28n^2 = b_n^2 - 2b_n$. Hence b_n is again even and we write $b_n = 2c_n$. The equation becomes $7n^2 = c_n(c_n - 1)$. Since $(c_n, c_n - 1) = 1$. We have two possibilities: (1) $c_n = A^2, c_n - 1 = 7B^2$; or (2) $c_n = 7A^2, c_n - 1 = B^2$. Finally, since $k^2 \equiv 0, 1, 4 \pmod{7}$. Case (2) is not possible. Thus we have $a_n = 4c_n = (2A)^2$ as claimed. $\square$

4.3 Mod n

The technique we now discuss base on the following easy principle:

“If an equation is not solvable mod n for some positive integer n , then it is not solvable in integers.”

The reason for this is simply that the contrapositive:

“If an equation is solvable in integers, then it is solvable mod n for all n ” is trivially true.

Example 4.8. Prove that the equation $y^2 = 3x - 1$ has no solution in integers.

Solution. Considering the equation mod 3 yields $y^2 \equiv 2 \pmod{3}$. This has no solution. Hence the original equation has no solution in integers. $\square$

Example 4.9. Prove that there are no integers x, y such that $x^3 + y^3 = 2006$.

Proof. Consider mod 7. We see that $a^3 \equiv 0, 1, -1 \pmod{7}$. Now $2006 \equiv 4 \pmod{7}$. Hence the equation $x^3 + y^3 \equiv 4 \pmod{7}$ is not solvable. Thus the original equation is not solvable in integers. $\square$

Example 4.10. Show that $x^2 + y^2 - 5xy = 5$ is not solvable in integers.

Proof. Completing the square gives $(x - y)^2 - 3xy = 5$. Now consider mod 3, (Why? Just because we can kill the term $3xy$.) this gives $(x - y)^2 \equiv 2 \pmod{3}$. This is not solvable. Hence the original equation has no solution.

Alternatively, if one choose to complete the square in this way: $(x + y)^2 - 7xy = 5$. Then mod 7 also yields no solution. $\square$

Example 4.11. Prove that the equation $x^2 + 3xy - 2y^2 = 122$ has no integer solutions.

Proof. We complete the square: $(x + 3y/2)^2 - 9y^2/4 - 2y^2 = 122$. Multiplying both sides by 4 yields: $(2x + 3y)^2 - 17y^2 = 488$. We next consider mod 17 give: $(2x + 3y)^2 \equiv 12 \pmod{17}$. This is not possible because $a^2 \equiv 0, 1, 2, 4, 8, 11, 13, 15, 16 \pmod{17}$.

Remark: Using Quadratic Reciprocity, which we will discuss later. It is very easy to determine whether a number is a square mod a prime p . $\square$

Example 4.12. Prove that $y^2 = x^3 + 7$ has no solution in integers.

Proof. First consider the **parity** of x . If x is even then we have $y^2 \equiv 3 \pmod{4}$, which is absurd. Hence x is odd. If $x \equiv 3 \pmod{4}$, then $y^2 \equiv 2 \pmod{4}$, which is absurd and thus $x \equiv 1 \pmod{4}$. Now write the equation as: $y^2 + 1 = x^3 + 8 = (x + 2)(x^2 - 4x + 4)$. Since $x + 2 \equiv 3 \pmod{4}$. There exists a prime $p \equiv 3 \pmod{4}$ and p divides $x + 2$. This implies $y^2 + 1 \equiv 0 \pmod{p}$, and this has no solution since -1 is a square mod p iff $p \equiv 1 \pmod{4}$. Therefore the original equation has no solution in integers. $\square$

Question: How do one decide which n to choose?

Answer: No one knows. Picking the right choice of n requires a combination of insight, experience, and luck. However, there are certain principles, which we will explain here.

Theorem 4.3 (Lagrange). *Fix a positive integer n , and an integer a such that $(a, n) = 1$. Let k be the smallest integer such that $a^k \equiv 1 \pmod{n}$, then k divides $\varphi(n)$.*

Proof. This is problem 3.6. For those who are familiar with basic group theory, this follows directly from the Lagrange's theorem in group theory (that's why we call it Lagrange's theorem here) that the order of the subgroup of a finite group G divides the order of G . $\square$

Theorem 4.4 (Cubing lemma). *If $p \equiv 2 \pmod{3}$, then every number a is a cube mod p , while if $p \equiv 1 \pmod{3}$ then exactly $(p - 1)/3 + 1$ are cubes mod p .*

Proof. We will use the fact that for a prime p , there exists g such that for all a , we have $a \equiv g^k \pmod{p}$ for some $0 \leq k \leq p - 2$, furthermore $g^k \equiv 1 \pmod{p}$ if and only if $p - 1 \mid k$. (In group theoretic language, this means $\mathbb{F}_p^\times$ is a cyclic group). We consider the equation

$x^3 \equiv a \pmod{p}$. First assume $p \equiv 2 \pmod{3}$, write $x \equiv g^n \pmod{p}$, $a \equiv g^k \pmod{p}$. Then we need to solve $g^{3n} \equiv g^k \pmod{p}$. i.e.

$$g^{3n-k} \equiv 1 \pmod{p}.$$

Hence we have $p-1 \mid 3n-k$ i.e. $3n \equiv k \pmod{p-1}$. But since $(3, p-1) = 1$, this is always solvable and thus every a is a cube when $p \equiv 2 \pmod{3}$. Now if $p \equiv 1 \pmod{3}$. We have to solve $x^3 \equiv a \pmod{p}$. As before, it is equivalent to solve $3n \equiv k \pmod{p-1}$. This is solvable iff $d = (3, p-1) = 3 \mid k$. Hence a is a cube iff $3 \mid k$. Now we ask how many such k 's are there? The answer is $(p-1)/3$ of which are non-zero, and $k = 0$ totalling in $(p-1)/3 + 1$. Done. $\square$

Remark: For those who are familiar with group theory, this theorem can be stated more elegantly as saying the cubing map $\varphi : \mathbb{F}_p^\times \rightarrow \mathbb{F}_p^\times$, where $x \mapsto x^3$ is an automorphism if $p \equiv 2 \pmod{3}$ and $|\ker(\varphi)| = 3$ if $p \equiv 1 \pmod{3}$. We shall also give a proof of the above theorem in this context. It is enough to show that $\ker(\varphi) = 1$ when $p \equiv 2 \pmod{3}$ and $|\ker(\varphi)| = 3$ when $p \equiv 1 \pmod{3}$.

Now, $\ker(\varphi) = \{x \in \mathbb{F}_p^\times \mid x^3 = 1\}$. If $x \neq 1$, then $x^3 = 1$ implies $3 \mid p-1$ by Lagrange's theorem. i.e. $p \equiv 1 \pmod{3}$. Hence if $p \equiv 2 \pmod{3}$, then $x^3 = 1$ iff $x = 1$, while if $p \equiv 1 \pmod{3}$, then there exists $x \neq 1$ (namely $x = g^{(p-1)/3}$ such that $x^3 = 1$ and this equation has three solutions namely $1, g^{(p-1)/3}, g^{2(p-1)/3}$. This concludes the proof. If one dislike the proof using the fact that $\mathbb{F}_p^\times$ is cyclic, one could also use quadratic reciprocity since $x^3 = 1$ has solution other than $x = 1$ iff $x^2 + x + 1 = 0$. This is true iff the Legendre symbol $(-3/p) = 1$ iff $p \equiv 1 \pmod{3}$.

The cubing lemma suggests that if we want to restrict the choices of $x^3 \pmod{p}$, we should choose a prime $p \equiv 1 \pmod{3}$. In general, if one wants to restrict the choices of $x^q \pmod{p}$, it is best to choose $p \equiv 1 \pmod{q}$.

Example 4.13. Show that $y^2 = x^5 + 7$ has no solution in integers.

Proof. 5-th power lemma suggests that we should first try to mod 11, since it's the first prime that is $1 \pmod{5}$. The $\{x^5 \mid x \in \mathbb{F}_{11}\}$ has size $1 + (11-1)/5 = 3$. Namely $0, \pm 1$ are the only 5-th powers modulo 11. This says $y^2 \equiv 7, 8, 6 \pmod{11}$. It is now a straight forward exercise to check the Legendre symbol calculations: $(7/11) = (8/11) = (6/11) = -1$, and hence the equation has no solution mod 11, and therefore no integer solution. $\square$

4.4 Primitive Pythagorean Triples

Theorem 4.5. *Primitive Pythagorean Triples Suppose $x^2 + y^2 = z^2$, and $(x, y, z) = 1$. Then:*

$$x = m^2 - n^2, y = 2mn, z = m^2 + n^2$$

for some m, n with $(m, n) = 1$ and $m \not\equiv n \pmod{2}$.

Proof. Suppose z is even, then $z^2 \equiv 0 \pmod{4}$. Assuming not both x, y are even, then they must be both odd. However, $x^2 \equiv 1 \pmod{4}$ for x odd. Hence $x^2 + y^2 \equiv 2 \pmod{4}$. This is a contradiction. Hence z must be odd and we may WOLOG assume that x is odd and y is even. Rewrite the equation as: $y^2 = z^2 - x^2 = (z - x)(z + x)$.

Let $d = (z - x, z + x)$. Then $d \mid 2z$, and $d \mid 2y$. Since $(x, z) = 1$, we have $d \mid 2$. Since, $z - x$ and $z + x$ are both even hence $d = 2$. This $z + x = 2m^2$, and $z - x = 2n^2$. Solving for x, y , and z gives the desired result at once. $\square$

4.5 Fermat-Pell's Equation: $x^2 - Dy^2 = 1$

Find three numbers that are simultaneously a square and a triangular number. Can you find more? Are there infinitely many such numbers?

This is a very naive problem and a simple check finds: 1, and 36 work. The question is: Are there more? Can we find them all?

The situation is not very clear and it seems pointless by random trial. Let us try to find satisfactory answer to these questions.

First of all, the n th triangular number $T(n)$ is equal to $1 + 2 + \dots + n = n(n+1)/2$. Hence we need to solve:

$$\frac{n(n+1)}{2} = m^2$$

in positive integers.

To this end, we rewrite the equation as $n(n+1) = 2m^2$. Since $(n, n+1) = 1$ we have either $n = y^2, n+1 = 2x^2$ or $n = 2y^2, n+1 = x^2$. Subtracting gives: $x^2 - 2y^2 = \pm 1$.

Hence the question is now reduced to find integer solutions to the above equations. The question is: How many solutions are there?

If you try to play around with the equation $x^2 - 2y^2 = 1$ for a little bit, you will see that after the pair $(x, y) = (17, 12)$, you won't find any until the next pair $(x, y) = (99, 70)$, and you might suspect there is no more solutions. The fact is, there are infinitely many solutions to this equation. It's just that the "size" of the solution grows very quickly. To this end, we have the following theorem:

Theorem 4.6 (Pell's Equation). *The equation:*

$$x^2 - 2y^2 = 1$$

has infinitely many solutions in positive integers.

All the solutions are obtained in the following way: write $(3 + 2\sqrt{2})^n = x_n + y_n\sqrt{2}$, then $x_n^2 - 2y_n^2 = 1$. Furthermore, all solutions arise in this way.

Proof. First of all, if $(a, b), (c, d)$ are solutions. Then consider the numbers $a + b\sqrt{2}, c + d\sqrt{2}$. Consider the following factorization: $(a + b\sqrt{2})(a - b\sqrt{2}) = 1$, it is clear that if $(a + b\sqrt{2})(c + d\sqrt{2}) = x + y\sqrt{2}$, then (x, y) is again a solution. Hence “products of solutions is again a solution”. Now, it is clear that the number $3 + 2\sqrt{2}$, generated by $(3, 2)$ is the smallest solution such that $x^2 - 2y^2 = 1$. Now, for each solution (a, b) , consider the number $a + b\sqrt{2}$, and we suppose that $a + b\sqrt{2}$ is not a power of $3 + 2\sqrt{2}$. We will derive a contradiction using infinite descent. Since $a + b\sqrt{2}$ is not a power of $3 + 2\sqrt{2}$, it must be trapped between two consecutive powers of $3 + 2\sqrt{2}$. In other words, there exists n such that:

$$(3 + 2\sqrt{2})^n < a + b\sqrt{2} < (3 + 2\sqrt{2})^{n+1}.$$

Now, multiply the inequality by $(3 - 2\sqrt{2})^n$, and note that this is a positive number. This yields:

$$1 < (a + b\sqrt{2})(3 - 2\sqrt{2})^n < 3 + 2\sqrt{2}.$$

Note that $(a + b\sqrt{2})(3 - 2\sqrt{2})^n = x + y\sqrt{2}$ is another solution because products of solutions is again a solution. We also note that x, y are necessarily positive. First of all, x, y cannot both be negative for $x + y\sqrt{2} > 1$. Hence either x or y is positive. We consider the two cases where exactly one of them is positive:

- (1) $x < 0, y > 0$.
- (2) $x > 0, y < 0$.

For (1), since $1 = x^2 - 2y^2 = (x + y\sqrt{2})(x - y\sqrt{2})$, and $x < 0, y > 0$ implies $x + y\sqrt{2} > 0, x - y\sqrt{2} < 0$ and thus $1 = (x + y\sqrt{2})(x - y\sqrt{2}) < 0$, a contradiction.

For (2), we have $x - y\sqrt{2} > x + y\sqrt{2}$ and since $1 = x^2 - 2y^2 = (x + y\sqrt{2})(x - y\sqrt{2})$ we must have $x - y\sqrt{2} > 1$ and $x + y\sqrt{2} < 1$. But this contradicts that fact that we have: $1 < x + y\sqrt{2} < 3 + 2\sqrt{2}$. Hence x, y are both positive. Furthermore, $x + y\sqrt{2}$ is a solution smaller than $3 + 2\sqrt{2}$. This contradicts the minimality of $3 + 2\sqrt{2}$.

Therefore, all solutions are generated by $(3 + 2\sqrt{2})^n$. □

We now cite a few results that are used frequently:

Theorem 4.7.

- (1) The solutions to $x^2 - 2y^2 = -1$ are generated by $x + y\sqrt{2} = (1 + \sqrt{2})^{2n+1}$.
- (2) The solutions to $x^2 - 3y^2 = 1$ are generated by $(2 + \sqrt{3})^n$.
- (3) The solutions to $x^2 - 5y^2 = \pm 1$ are generated by $x + y\sqrt{5} = \pm(2 + \sqrt{5})^n = \pm\alpha^{3n}$, where $\alpha = (-1 + \sqrt{5})/2$, the parity of n depends on the parity of the equation.
- (4) The solutions to $x^2 - 5y^2 = \pm 4$ are generated by $(x + \sqrt{5})/2 = \alpha^n$, the parity of n depends on the parity of the equation.

Theorem 4.8. Let $D > 0$ be a square free integer. Then $x^2 - Dy^2 = 1$ has infinitely many integer solutions. Let (u, v) be a minimal solution to $x^2 - Dy^2 = c$, and (a, b) be

a minimal solution to $x^2 - Dy^2 = 1$, then all solutions of $x^2 - Dy^2 = c$ are given by $(u + v\sqrt{D})(a + b\sqrt{D})^n = x_n + y_n\sqrt{D}$.

Example 4.14 (Putnam 2000). There are infinitely n , such that $n, n+1, n+2$ are each the sum of two integer squares. [Example: $0 = 0^2 + 0^2, 1 = 1^2 + 0^2, 2 = 1^2 + 1^2$.]

Proof. The Pell's equation $x^2 - 2y^2 = 1$ has infinitely many integer solutions. Now let $n = 2y^2 = y^2 + y^2, n+1 = x^2 = x^2 + 0^2, n+2 = x^2 + 1^2$. Done. $\square$

Example 4.15. Show that there are infinitely many positive integer solutions to $a^2 + b^3 = c^4$.

Proof. One way to attack such question is simply find one solution (a_0, b_0, c_0) and then $(a_0 t^6, b_0 t^4, c_0 t^3)$ as t runs through the integers will give an infinite family of solutions.

Alternatively, we use the fact:

$$1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2 = \left[\frac{n(n+1)}{2} \right]^2.$$

Set $a = (n-1)n/2, b = n^3$, we have $a^2 + b^3 = [n(n+1)/2]^2$. It suffices to show that $n(n+1)/2$ is a perfect square for infinitely many n . This is equivalent to show that there are infinitely many triangular numbers that are also square numbers, and which is equivalent to say that $x^2 - 2y^2 = \pm 1$ has infinitely many integer solutions as asserted at the beginning of the section. Done. $\square$

4.6 When Size Matters

Example 4.16. Determine all pairs (x, y) of positive integers satisfying: $(x+y)^2 - 2(xy)^2 = 1$.

This might strike you as a Pell's equation problem, however, you will soon realize how hopeless it is to apply theory about Pell's equation. The key idea is that we expect the product xy grows a lot faster than $x+y$ and thus we expect that if both x, y are large, then the left hand side is negative.

Proof. Suppose $x, y \geq 2$. Then $x - x/y \geq x - x/2 = x/2 \geq 1$. Rearranging gives $xy \geq x+y$. Now, $\sqrt{2}xy > xy \geq x+y$. Squaring gives $2(xy)^2 \geq (x+y)^2$. This says the equation is not solvable if $x, y \geq 2$. Since the equation is symmetric in x, y , we may WOLOG assume $x = 1$. Expanding and simplifying gives $y^2 + 2y = 0$, which clearly is not solvable for positive integer y . Hence the equation is not solvable. $\square$

Example 4.17. Solve $a^2 + b^2 - ab = 1$ in integers.

Proof. This is equivalent to $(2a - b)^2 + 3b^2 = 4$. Hence we have either $|2a - b| = 2, b = 0$ or $|2a - b| = 1, |b| = 1$. In the first case, we have $a = \pm 1, b = 0$. In the second case, we have $(a, b) = (0, -1), (0, 1), (1, 1), (-1, -1)$. Done. $\square$

Example 4.18 (Putnam 2005). Find all positive integers $n, k_1, \dots, k_n$ such that $k_1 + \dots + k_n = 5n - 4$ and

$$\frac{1}{k_1} + \dots + \frac{1}{k_n} = 1.$$

Proof. By AM-HM inequality, $(5n - 4)/n \geq n/1$, which implies $5n - 4 \geq n^2$. This is equivalent to $n^2 - 5n + 4 = (n - 4)(n - 1) \leq 0$, which means $1 \leq n \leq 4$. This implies $n = 1, 2, 3$ or 4 . For $n = 1$, we have $k_1 = 1$. For $n = 2$, we have $k_1 + k_2 = 6$ and $1/k_1 + 1/k_2 = (k_1 + k_2)/(k_1 k_2) = 1$. This implies $k_1 k_2 = 6$, hence k_1, k_2 are solutions to $x^2 - 6x + 6 = 0$, which yields no integer solution. For $n = 3$, we need to solve $k_1 + k_2 + k_3 = 11$, and $1/k_1 + 1/k_2 + 1/k_3 = 1$. WOLOG assume that $k_1 \geq k_2 \geq k_3 > 1$. If $k_3 = 2$, then we have $k_1 + k_2 = 9, k_1 k_2 = 18$. Hence $(k_1, k_2) = (6, 3)$ is the unique solution. If $k_3 = 3$, then we have $k_1 + k_2 = 8, k_1 k_2 = 24$, which yields no integer solution. Finally, note that $k_3 < 4$ otherwise $k_1, k_2, k_3 \geq 4$ implies $1/k_1 + 1/k_2 + 1/k_3 < 3/4 < 1$. Therefore we have found all solutions for $n = 3$, viz. $(k_1, k_2, k_3) = (6, 3, 2), (6, 2, 3), (3, 6, 2), (3, 2, 6), (2, 3, 6), (2, 6, 3)$. Finally, for $n = 4$, we have $(5n - 4)/n = 16/4 = n = 4$, therefore equality condition of AM-HM gives $k_1 = k_2 = k_3 = k_4 = 4$. Done. $\square$

Example 4.19. Solve $x^3 - y^3 = xy + 61$ in positive integers.

Proof. It is clear that $x > y$, write $x = y + d, d \geq 1$, then we have:

$$x^3 - y^3 - xy - 61 = (3d - 1)y^2 + d(3d - 1)y + d^3 - 61 = 0.$$

Since $d \geq 1$, we have $3d - 1 > 0$. This implies $d^3 - 61 < 0$ otherwise all terms on the left are strictly positive. This says $d = 1, 2$, or 3 . For $d = 1$, we have $2y^2 + 2y - 60 = 0$ which is equivalent to $y^2 + y - 30 = (y - 6)(y + 5) = 0$. Therefore $(x, y) = (7, 6)$ is a solution. For $d = 2$, we have $5y^2 + 10y - 53 = 0$, which yields no solution. Finally, for $d = 3$, we have $8y^2 + 24y - 35 = 0$, which yields no solution. Therefore the unique solution is $(x, y) = (7, 6)$. $\square$

Example 4.20. Solve $x^y = y^x$ in positive integers, such that $x > y$.

Proof. Rewrite the equation as $x^{1/x} = y^{1/y}$. Let $f(t) = t^{1/t}$. Logarithmic differentiation yields: $f'(t) = f(t)(1 - \log(t))/t^2 > 0$ iff $t < e$. Hence, $f(t)$ is increasing in the interval $[1, e)$ and decreasing for $t > e$. Thus if $y \geq 3$, then $f(y) > f(x)$. Hence $y \leq 2$. Only $y = 2$ yields the solution $(x, y) = (4, 2)$, satisfying the assumption $x > y$. Done. $\square$

Example 4.21. Given that a rectangular box with dimension a, b, c has the property that when each length, width, and depth is increased by 1 unit. The new volume is twice the original volume. Find all possible such boxes.

Proof. We want to solve $(a+1)(b+1)(c+1) = 2abc$ in positive integers. It is easy to see that if $a = 1$, then there is no solution. So we assume $1 < a \leq b \leq c$.

Look at $f(a, b, c) = [(a+1)(b+1)(c+1)]/(abc) = (1+1/a)(1+1/b)(1+1/c)$. If one expands the product, it is easy to see that f is decreasing. If $a \geq 4$, then $1 < f(a, b, c) \leq f(4, 4, 4) = 125/64 < 2$. Hence $f(a, b, c)$ cannot be 2 for $a \geq 4$. Hence $a = 2$, or 3.

Case (1): If $a = 2$, then we solve $3(b+1)(c+1) = 4bc$. This says $3 \mid b$ or $3 \mid c$. By symmetry, we may WOLOG assume $3 \mid b$, and write $b = 3k$. Then we have $(3k+1)(c+1) = 4kc$. Now $(k, 3k+1) = (c, c+1) = 1$ implies $k \mid c+1$ and $c \mid 3k+1$. Writing $c+1 = km$, $3k+1 = cn$ gives $mn = 4$. i.e. $(m, n) = \{(1, 4), (4, 1), (2, 2)\}$. We now study each of these cases carefully:

If $(m, n) = (1, 4)$, then $c+1 = k$, $3k+1 = 4c$. Solving gives $3k+1 = 4(k-1)$ i.e. $k = 5$, $c = 4$. This yields $b = 15$. i.e. $(a, b, c) = (2, 15, 4)$. For convention, to keep the condition $1 < a < b < c$. We reorder to get $(a, b, c) = (2, 4, 15)$.

If $(m, n) = (4, 1)$, then $c+1 = 4k$, $3k+1 = c$. Solving gives $3k+1 = 4k-1$. i.e. $k = 2$, $c = 7$. This yields $b = 6$. i.e. $(a, b, c) = (2, 6, 7)$.

If $(m, n) = (2, 2)$, then $c+1 = 2k$, $3k+1 = 2c$. Solving gives $3k+1 = 2(2k-1)$. i.e. $k = 3$, $c = 5$. This yields $(a, b, c) = (2, 9, 5)$, and again we reorder to get $(2, 5, 9)$.

Case (2): Now if $a = 3$, we need to solve $4(b+1)(c+1) = 6bc$, or $2(b+1)(c+1) = 3bc$. WOLOG, assume $b = 2k$. i.e. $(2k+1)(c+1) = 3kc$. Again, $(k, 2k+1) = (c, c+1) = 1$ gives $k \mid c+1$, $c \mid 2k+1$. Write $c+1 = km$, $2k+1 = cn$ yields $mn = 3$. i.e. $(m, n) = \{(1, 3), (3, 1)\}$.

If $(m, n) = (1, 3)$, then we solve $2k+1 = 3c = 3(k-1)$. i.e. $k = 4$, $c = 3$. This gives $(a, b, c) = (3, 8, 3)$ and reorder to get $(3, 3, 8)$.

If $(m, n) = (3, 1)$, then we solve $2k+1 = c = 3k-1$. i.e. $k = 2$, $c = 5$. This gives $(a, b, c) = (3, 4, 5)$.

Hence the solutions are: $(a, b, c) = \{(2, 4, 15), (2, 6, 7), (2, 5, 9), (3, 3, 8), (3, 4, 5)\}$. Done. $\square$

4.7 Fermat's Infinite Descent

Theorem 4.9. $x^4 + y^4 = z^2$ has no nontrivial solutions in integers.

Proof. We may first assume that x, y and z are relatively prime. Among all positive integer solutions (x, y, z) we pick one such that z is positive and minimal. Then using the solution to primitive Pythagorean triple, write $x^2 = m^2 - n^2$, $y^2 = 2mn$, and $z = m^2 + n^2$. Here, m, n are relatively prime and of opposite parity. Now, $x^2 + n^2 = m^2$ and $(x, m, n) = 1$

implies $x = s^2 - t^2, m = 2st$ and $n = s^2 + t^2$, where s, t are relatively prime and of opposite parity. Look at $y^2 = 2mn$ and we have: $y^2 = 2(2st)(s^2 + t^2) = 2^2 st(s^2 + t^2)$. It is easy to see that $s, t, s^2 + t^2$ are pairwise relatively prime and thus each of s, t and $s^2 + t^2$ is a square. Write $s = a^2, t = b^2$, and $s^2 + t^2 = c^2$. This implies $a^4 + b^4 = c^2$ and $c = \sqrt{s^2 + t^2} = \sqrt{n} < m^2 + n^2 = z$. This contradicts the minimality of z . Therefore, the equation has no positive integer solutions. $\square$

Corollary 4.10 (Fermat's Last Theorem for $n = 4$). $x^4 + y^4 = z^4$ has no nontrivial solution in integers.

Proof. Leave as an exercise. $\square$

Proposition 4.11. The diophantine equation $x^4 + 2y^4 = z^2$ has no solution with $y \neq 0$.

Proof. WOLOG, we may assume that x, y, z are pairwise relatively prime. We rewrite the equation as $(z - x^2)(z + x^2) = 2y^4$ and let $d = (z - x^2, z + x^2)$. Then $d \mid 2z$, and $d \mid 2x^2$. Since $(x, z) = 1$, we have $d = 1$ or 2 . However, since $x \equiv z \pmod{2}$, we have $d = 2$. This implies either $z - x^2 = 2a^4, z + x^2 = b^4$, where a odd, b even or $z - x^2 = a^4, z + x^2 = 2b^4$, where a even, b odd.

In the first case, we have $2x^2 = b^4 - 2a^4$, which is $x^2 = b^4/2 - a^4$. Since x and a are odd, and $8 \mid b^4/2$. We have $1 \equiv -1 \pmod{8}$ which is a contradiction.

In the second case, we have $b^4 - x^2 = 8A^4$, where $a = 2A$. We again have $(b^2 - x, b^2 + x) = 2$. Hence either $b^2 - x = 2u^4, b^2 + x = 4v^4$ or $b^2 - x = 4u^4, b^2 + x = 2v^4$. This yields either $b^2 = u^4 + 2v^4$ or $b^2 = v^4 + 2u^4$, and we are done by descent. $\square$

Example 4.22. Show that $x^2 + y^2 + z^2 = 6xyz$ has no integer solution other than $x = y = z = 0$.

Analysis. Considering mod 2 implies that x, y, z cannot be all odd. If x is even, then considering mod 4 gives: $y^2 + z^2 \equiv 0 \pmod{4}$. This implies y, z are also even. Writing x, y, z as $2a, 2b, 2c$ respectively and canceling common powers of 2 on both sides give:

$$a^2 + b^2 + c^2 = 12abc.$$

Applying the exact same analysis says a, b, c are again all even. This suggests that x, y, z are “infinitely” divisible by 2, which is clearly impossible. This suggests that infinite descent should work. $\square$

Proof. If any one of x, y, z is equal to zero, then it is easy to see that $x = y = z = 0$. Hence, we assume that x, y, z are non-zero.

Let $x = 2^a A, y = 2^b B, z = 2^c C$, where A, B and C are odd. Since the expression is symmetric in x, y, z , we may WOLOG assume that $a \geq b \geq c$. Then the original equation becomes:

$$2^{2a}A^2 + 2^{2b}B^2 + 2^{2c}C^2 = 2^{a+b+c+1}3ABC.$$

Factoring 2^{2c} on both sides give:

$$2^{2a-2c}A^2 + 2^{2b-2c}B^2 + C^2 = 2^{a+b-c+1}3ABC.$$

Now, we consider cases:

- (1) $a > b > c$, then left hand side is odd while the right hand side is even.
- (2) $a > b = c$, then right hand side is $0 \pmod{4}$ implies $B^2 + C^2 \equiv 0 \pmod{4}$. This says B and C are even, a contradiction to our assumption that B, C are odd.
- (3) $a = b > c$, considering $\pmod{4}$ yields the same contradiction as in (2).
- (4) $a = b = c$, considering $\pmod{2}$ yields the same contradiction as in (1).

Hence we conclude that the equation has no non-zero integer solutions.

□

4.8 IMO 1988 Problem 6

The main character of this section is the following infamous problem from IMO 1988. This is one of the most difficult problems in the IMO history. In this section, we will discuss its solution and many other problems whose solution uses the same technique.

Example 4.23 (IMO 1988 Q6). Suppose a, b are positive integers and $k = (a^2 + b^2)/(ab + 1)$ is a positive integer. Prove that k is a perfect square.

Proof. This solution is based on a contestant's solution, which won the special prize of this year's IMO. Suppose k is not a square, then $k > 2$. Since the expression is symmetric in a, b , we may WOLOG assume $a \geq b$ and if $a = b$, then $k = (2 - k)a^2 < 0$. Hence we must have $a > b$. Now, among all the pairs (a, b) such that k is a positive integer. Choose a pair such that $a + b$ is minimal. Consider the quadratic equation:

$$x^2 - kbx + b^2 - k = 0.$$

First note that a is a root, and we let a' be the other root. Then by Viète's theorem, $a + a' = kb$ and $aa' = b^2 - k$. This implies that a' is an integer.

First claim that $a' > 0$, this is so because $k(1 + a'b) = (a')^2 + b^2 > 0$ and $b > 1$. Hence if $a' < 0$, then $1 + a'b < 0$ and we would have a contradiction.

Next claim that $a' < a$. This is so because $a' = (b^2 - k)/a < b^2/a < a$.

Hence (a', b) is another pair satisfying $x^2 + b^2 - xbk = k$ such that $a' + b < a + b$. This is a contradiction.

Hence k is a square.

□

The following theorem is a generalization of this problem:

Theorem 4.12. *Suppose a, b, c are positive integers such that $0 < a^2 + b^2 - abc \leq c$. Then $a^2 + b^2 - abc$ is a perfect square.*

Proof. Let $k = a^2 + b^2 - abc$, and suppose (a, b) is such that $a + b$ is minimal. Suppose k is not a square, then $k > 2$. Since the expression is symmetric in a, b , we may WOLOG assume $a \geq b$ and if $a = b$, then $k = (2 - c)a^2 \leq (2 - k)a^2 < 0$. Hence we must have $a > b$.

Consider the equation: $x^2 - (bc)x + b^2 - k = 0$, let a, a' be the roots. Then we have $a + a' = bc$, thus a' is an integer. We also have $aa' = b^2 - k$.

Claim 1 $a' > 0$

Check. If $a' < 0$, then $1 + a'b < 0$ since $b > 1$. Thus $0 > c(1 + a'b) = c + a'bc \geq k + a'bc = (a')^2 + b^2$. This is a contradiction. Hence we have $a' > 0$ as claimed. $\square$

Claim 2 $a' < a$.

Check. $a' = (b^2 - k)/a < b^2/a < a$. $\square$

Hence (a', b) is another pair satisfying $x^2 + y^2 - xyc = k$ such that $a' + b < a + b$. This is a contradiction.

Hence k is a square. $\square$

Example 4.24 (Estonian MO 2000). Suppose x, y are positive integers such that $xy \mid x^2 + y^2 + 6$. Find all possible values of $(x^2 + y^2 + 6)/xy$.

Proof. For a fix k , among all positive integers x, y such that $(x^2 + y^2 + 6)/xy = k$. Choose a pair such that $x + y$ is minimal. Since the expression is symmetric in x, y , we may WOLOG assume that $x \geq y$. If $x = y$, then we have $x^2 \mid 6$ and hence $x = 1$. This gives $k = 8$.

Now assume $x > y$. Consider $x^2 - kxy + y^2 + 6 = 0$ as a quadratic in x and let x' be its other root. Then by Viète's theorem, we have $xx' = y^2 + 6$. This implies $x' > 0$. Since $x > y$, we have $x \geq y + 1$. Now, $x' = (y^2 + 6)/x \leq [(x - 1)^2 + 6]/x = (x^2 - 2x + 5)/x$. If $x \geq 3$, then $(x^2 - 2x + 5)/x < x$. Hence $x' < x$, and therefore $x' + y < x + y$ and this contradicts the minimality of $x + y$. Thus $x \leq 2$. Therefore, $2 \geq x \geq y + 1$, which implies $y = 1$.

If $y = 1$, then $x \mid x^2 + 7$ and thus $x \mid 7$. Thus $x = 7$ and $k = 8$. Therefore, $k = 8$ is the unique value. $\square$

Example 4.25. Suppose a, b, c are positive integers such that $a^2 + b^2 + c^2 = kabc$, where k is a positive integer. Find all possible values of k .

Proof. The strategy is to first control the size of k by giving it a bound. Due to symmetry, we may WOLOG assume that $a \geq b \geq c$. Among all triples (a, b, c) that solves the diophantine equation, we choose one such that $a + b + c$ is minimal. We can also view a as a root of the quadratic: $x^2 - kbcx + b^2 + c^2 = 0$ and let the other root be a' . In this case, we have $a' \geq a$, for if otherwise, we would have contradicted the minimality of $a + b + c$.

Now, we have:

$$k = \frac{a^2 + b^2 + c^2}{abc} = \frac{a}{bc} + \frac{b}{ca} + \frac{c}{ab} \leq \frac{a}{bc} + 1 + 1.$$

We consider two cases:

- (1) $a \geq bc$ and $a' \geq bc$.
- (2) $a < bc$ or $a' < bc$.

For case (1), by Viète's theorem, we have $a + a' = kbc$, and $aa' = b^2 + c^2$. Hence, $b^2 + c^2 = aa' \geq a^2 \geq (bc)^2$. This implies: $c^2 \geq b^2(c^2 - 1)$. Therefore, if $c > 1$, we would have $c^2 > b^2$, which is a contradiction. Hence $c = 1$. Now, we have $a + a' = kb$, $aa' = b^2 + 1$. If $a = b$, then we have $k = (2a^2 + 1)/a^2 = 2 + 1/a^2$, this implies $a = 1$, which yields $k = 3$. So, we may now assume $a > b$, thus $a \geq b + 1$. In this case, $a' = (b^2 + 1)/a \leq [(a - 1)^2 + 1]/a = (a^2 - 2a + 2)/a < a$ if $a > 1$, thus $a = 1$. This implies $b < 1$, which is a contradiction. Therefore, there is no solution other than $a = b = c = 1$, and this gives $k = 3$.

For case (2), since $a' \geq a$, we must have $a < bc$. Therefore, $k < 3$. For $k = 1$, $a = b = c = 3$ is a solution. We will show that $k = 2$ yields no solution other than the trivial solution $a = b = c = 0$. To see this, write $a = 2^p A$, $b = 2^q B$, $c = 2^r C$, where A, B, C are odd. WOLOG, we may assume $p \geq q \geq r$. Then, we have: $2^{2r}(2^{2p-2r}A^2 + 2^{2q-2r}B^2 + C^2) = 2^{p+q+r+1}ABC$. Since $p+q+r+1 \geq 2r$. We can rewrite the equation as: $2^{2p-2r}A^2 + 2^{2q-2r}B^2 + C^2 = 2^{p+q-r+1}ABC$. If $p = 0$, then the left hand side is odd, while the right hand side is even. Hence $p \geq 1$, then the right hand side is 0 (mod 4), while the left hand side is either 1, 2 or 3 (mod 4). Hence the equation $a^2 + b^2 + c^2 = 2abc$ has no positive integer solution.

We therefore conclude that $k = 1$ or $k = 3$ are the only possible values. $\square$

4.9 Rational Parametrization of Conics

The following method of finding rational solutions to a conic is called rational parametrization.

Example 4.26. Find all rational solutions to $x^2 + y^2 = 1$.

Proof. It is easy to see that $(x, y) = (-1, 0)$ is such a solution. Now let $L : y = t(x + 1)$ be the line passes through the point $(-1, 0)$ with slope t . This line intersects the circle at an other point $P = (a, b)$. We equate the two equations to find the intersection:

$$x^2 + t^2(x+1)^2 = 1.$$

This simplifies to: $(t^2 + 1)x^2 + (2t^2)x + t^2 - 1 = 0$.

Since -1 is a root, by Viète's theorem, we have: $-a = (t^2 - 1)/(t^2 + 1)$. Hence we have

$$a = \frac{1 - t^2}{1 + t^2}, \quad b = \frac{2t}{1 + t^2}.$$

Next, we note that if $t \in \mathbb{Q}$, then a and b are rational. Conversely, if $a, b \in \mathbb{Q}$. Then $t \in \mathbb{Q}$ since $b = t(a + 1)$. Therefore, we have found all rational solutions. $\square$

Example 4.27. Find all rational solutions to $x^2 - 2y^2 = 1$.

Proof. We start with the rational point $(-1, 0)$. Let $L : y = t(x+1)$ be the line passes through the point $(-1, 0)$ with slope t as before. This line intersects the conic at an other point $P = (a, b)$. This yields $x^2 - 2t^2(x+1)^2 = 1$, which simplifies to: $(1 - 2t^2)x^2 - 2t^2x - 2t^2 - 1 = 0$. By Viète's theorem, $-a = -(t^2 + 1)/(1 - 2t^2)$, which implies $a = (t^2 + 1)/(1 - 2t^2)$, $b = t(a + 1)$ and we have found all rational solutions. $\square$

Remark: Provided that there is at least one rational point on the curve, this method works for finding all rational solutions of any conic, not just circles, or the Pell's type equation considered. For example, this won't work for $x^2 + y^2 = 3$ because there is no rational solution.

4.10 Squares in Fibonacci Sequence

In this section, we aim to classify all Fibonacci numbers that are also perfect squares.

Let α, β , with $\alpha > \beta$ be the two roots of $x^2 - x - 1 = 0$. Recall that the Fibonacci and Lucas numbers are defined as:

$$F_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}, \quad L_n = \alpha^n + \beta^n,$$

note that both F_n and L_n satisfy the recurrence $u_{n+2} = u_{n+1} + u_n$, with initial terms: $F_0 = 0, F_1 = 1, L_0 = 2, L_1 = 1$.

Lemma 4.13.

- (1) $L_{-n} = (-1)^n L_n$, $F_{-n} = (-1)^n F_n$, $L_n^2 - 5F_n^2 = 4(-1)^n$, $2L_{m+n} = 5F_m F_n + L_m L_n$, $2F_{m+n} = F_m L_n + F_n L_m$, $L_{2m} = L_m^2 + 2(-1)^{m-1}$, $F_{2m} = F_m L_m$.
- (2) $L_n \equiv 0 \pmod{2}$ iff $3 \mid n$, $F_n \equiv 0 \pmod{2}$ iff $3 \mid n$.
- (3) $\gcd(L_n, F_n) = 1$ if $3 \nmid n$, $\gcd(L_n, F_n) = 2$ if $3 \mid n$.

Proof. The formulas in (1) are direct consequences from the definition involving α and β . For (2), it is trivial since both sequences modulo 2 are periodic of period 3. For (3) since $|L_n^2 - 5F_n^2| = 4$, $\gcd(F_n^2, L_n^2) \mid 4$, which implies $(F_n, L_n) \mid 2$. By the exact formula, $(F_n, L_n) = 2$ iff F_n is even, and the assertion follows from (2). $\square$

Lemma 4.14. *When $k \equiv \pm 2 \pmod{6}$, then for all $t \in \mathbb{Z}$, we have:*

$$L_{n+2kt} \equiv (-1)^t L_n \pmod{L_k}, \quad F_{n+2kt} \equiv (-1)^t F_n \pmod{L_k}.$$

Proof. Since k is even, we have:

$$2L_{n+2k} = 5F_n F_{2k} + L_n L_{2k} = 5F_n F_k L_k + L_n (L_k^2 - 2) \equiv -2L_n \pmod{L_k}.$$

Since $3 \nmid k$, we have L_k is odd and hence $L_{n+2k} \equiv -L_n \pmod{L_k}$, and the assertion follows from induction on t .

Likewise, we have:

$$2F_{n+2k} = F_n L_{2k} + F_{2k} L_n = F_n (L_k^2 - 2) F_k L_k L_n \equiv -2F_n \pmod{L_k}$$

and we conclude as before. $\square$

Theorem 4.15 (Lucas Square). *For $n \geq 0$, we have:*

- (1) $L_n = x^2, x \in \mathbb{Z}$ iff $n = 1, 3$.
- (2) $L_n = 2x^2, x \in \mathbb{Z}$ iff $n = 0, 6$.

Proof. (1) Since $L_{2m} = L_m^2 + 2(-1)^{m-1}$, if $L_{2m} = x^2$, then $|x^2 - L_m^2| = 2$, a contradiction. Since L_1, L_3 are squares, we assume $n > 3$ and odd. Write $n = r + 2 \cdot 3^s k$, where $r = 1$ or 3 , and $2 \mid k, 3 \nmid k$. This implies $k \equiv \pm 2 \pmod{6}$, thus by lemma 4.14, we have:

$$L_n \equiv (-1)^{3^s} L_r \pmod{L_k}.$$

Since $L_1 = 1, L_3 = 4$, we have $L_n \equiv -1, -4 \pmod{L_k}$.

Next, $k/2 \equiv \pm 1 \pmod{3}$ implies $L_{k/2}$ is odd and hence:

$$L_k = L_{k/2}^2 + 2(-1)^{k/2-1} \equiv 1 \pm 2 \pmod{4}.$$

Therefore, we have the Jacobi symbol calculation:

$$\left(\frac{L_n}{L_k} \right) = \left(\frac{-L_r}{L_k} \right) = \left(\frac{-1}{L_k} \right) = -1,$$

where the last equality follows from the fact that $L_k \equiv 3 \pmod{4}$. This says that L_n is not a quadratic residue mod L_k and thus cannot be a square.

For (2), suppose $L_n = 2x^2$ and we consider the following cases:

- (a) If n is odd, then $4x^4 = L_n^2 = 5F_n^2 - 4$, which implies F_n is even. Hence $x^4 = 5(L_n/2)^2 - 1$. Now $x^4 \equiv 0, 1 \pmod{8}$ implies $(F_n/2)^2 \equiv 2, 5 \pmod{8}$, which is a contradiction.
- (b) If $4 \mid n \neq 0$, write $n = 2 \cdot 3^s k$, with $k \equiv \pm 2 \pmod{6}$ as before. By lemma 4.14,

$$2L_n \equiv -2L_0 \equiv -4 \pmod{L_k},$$

and we have the Jacobi symbol $(2L_n/L_k) = -1$ as before. This implies $2L_n$ cannot be a square.

- (c) If $n \equiv 6 \pmod{8}$ and $n \neq 6$, write $n = 6 + 2 \cdot 3^s k$ with $k \equiv \pm 2 \pmod{6}$. By lemma 4.14,

$$2L_n \equiv -2L_6 \equiv -36 \pmod{L_k}.$$

Therefore we also have $(2L_n/L_k) = -1$ provided that $(3, L_k) = 1$, which is true since $3 \mid L_m$ iff $m \equiv 2 \pmod{4}$, and we have $4 \mid k$.

- (d) If $n \equiv 2 \pmod{8}$ then $L_{-n} = L_n$ and $-n \equiv 6 \pmod{8}$ and case (c) says that $L_n = 2x^2$ iff $-n = 6$, and hence $n < 0$. $\square$

Theorem 4.16 (Fibonacci Square). *Let $n \geq 0$, we have:*

- (1) $F_n = x^2, x \in \mathbb{Z}$ iff $n = 0, 1, 2, 12$.
 (2) $F_n = 2x^2, x \in \mathbb{Z}$ iff $n = 0, 3, 6$.

Proof. (1) We consider the following cases:

- (a) $n \equiv 1 \pmod{4}, n \neq 1$ write $n = 1 + 2 \cdot 3^s k$, where $k \equiv \pm 2 \pmod{6}$. By lemma 4.14,

$$F_n \equiv (-1)^{3^s} F_1 \equiv -1 \pmod{L_k},$$

also, $L_k \equiv 3 \pmod{4}$ implies $(F_n/L_k) = (-1/L_k) = -1$ which implies F_n is not a quadratic residue mod L_k , hence not a square.

- (b) $n \equiv 3 \pmod{4}$, then $F_{-n} = F_n$ and $-n \equiv 1 \pmod{4}$ and hence F_n is not a square by the same argument as in (a).

- (c) $n \equiv 0 \pmod{6}, n \neq 0$, write $n = 6k$ and suppose $F_n = x^2$. Since $F_{6k} = F_{3k}L_{3k}$ and $(F_{3k}, L_{3k}) = 2$. This implies $L_{3k} = 2y^2$. However, by Lucas square theorem, this happens iff $3k = 0$ or $3k = 6$. Hence $n = 0$ or $n = 12$.

- (d) $n \equiv \pm 2 \pmod{6}$, write $n = 6k \pm 2$, and suppose $F_n = x^2$. Then we have $F_n = F_{3k \pm 1}L_{3k \pm 1}$. Since $(F_{3k \pm 1}, L_{3k \pm 1}) = 1$ We have $L_{3k \pm 1} = y^2$, which happens iff $3k \pm 1 = 1$ or 3 , by the Lucas square theorem. Hence $n = 2$.

For (2), we consider the following cases:

- (a) $n \equiv 3 \pmod{4}$, write $n = 3 + 2 \cdot 3^s k$, where $k \equiv \pm 2 \pmod{6}$, and in particular, $k \not\equiv 0$. By lemma 4.14,

$$F_n \equiv -F_3 \equiv -2 \pmod{L_k}.$$

As before, $L_k \equiv 3 \pmod{4}$ implies $(2F_n/L_k) = (-4/L_k) = -1$, and hence $2F_n$ is not a square, unless $k = 0$. (i.e. $n = 3$)

(b) $n \equiv 1 \pmod{4}$, we have $F_n = F_{-n}$ and since $-n \equiv 3 \pmod{4}$, it follows from (a) that $2F_n$ is not a square, unless $n = -3$.

(c) $n \equiv 0 \pmod{6}$, write $n = 6k$. Then we have $F_{6k} = F_{3k}L_{3k}$. Since $(F_{3k}, L_{3k}) = 2$, if $F_{6k} = 2x^2$. We have either $L_{3k} = y^2$ or $L_{3k} = 2z^2$. By the Lucas square theorem, the former case happens iff $3k = 0$ or $3k = 6$, and the latter case happens iff $3k = 1$ or $3k = 3$. We check that $F_{6k} = 2x^2$ iff $k = 0$ or $k = 1$.

(d) $n \equiv \pm 2 \pmod{6}$, write $n = 6k \pm 2$. Then we have $F_{6k \pm 2} = F_{3k \pm 1}L_{3k \pm 1}$, and $(F_{3k \pm 1}, L_{3k \pm 1}) = 1$. As in case (c), we invoke Lucas square theorem to obtain that $F_{6k \pm 2}$ is never twice a square. Done. □

4.11 Diophantine Approximation

Theorem 4.17 (Liouville). *Let α be an algebraic number of degree $n \geq 2$. Let $f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ such that $f(\alpha) = 0$ and that f is irreducible over $\mathbb{Q}$. Then for all $p/q \in \mathbb{Q}$, we have:*

$$\left| \alpha - \frac{p}{q} \right| \geq \frac{c(\alpha)}{q^n},$$

where $c(\alpha) = \min\{M, 1/|a_n|(3M)^{n-1}\}$. Here $M = \max\{|\alpha_i|\}$, where α_i are the roots of f .

Proof. We have $|f(\alpha) - f(p/q)| = |f(p/q)| \geq 1/q^n$ since the numerator of $f(p/q)$ is a non-zero integer. Now we consider two cases:

(1) $|p/q| > 2M$, then $|\alpha - p/q| \geq ||p/q| - |\alpha|| \geq M \geq M/q^n$, since $q \geq 1$.

(2) $|p/q| \leq 2M$, then $|\alpha_i - p/q| \leq |\alpha_i| + |p/q| \leq 3M$. We have $|a_n| \prod_{i=1}^n |\alpha_i - p/q| = |f(p/q)| \geq 1/q^n$. Let $\alpha_1 = \alpha$, we have

$$\left| \alpha - \frac{p}{q} \right| \geq \frac{1}{|a_n| \prod_{i=2}^n |\alpha_i - p/q| q^n} \geq \frac{1}{|a_n| (3M)^{n-1} q^n}.$$

Hence, we have

$$\left| \alpha - \frac{p}{q} \right| \geq \frac{c(\alpha)}{q^n}$$

as claimed. □

Example 4.28 (CGMO 2007). Let $a, b, c \in \mathbb{Z}$ such that $|a|, |b|, |c| \leq 10$. Let $f(x) = x^3 + ax^2 + bx + c$. Suppose $|f(2 + \sqrt{3})| < 10^{-4}$. Prove that $f(2 + \sqrt{3}) = 0$.

Proof. Let $\alpha = 2 + \sqrt{3}, \beta = 2 - \sqrt{3}$. Let $g(x) = x^2 - 4x + 1$, then $g(\alpha) = 0$ and f is clearly irreducible. (i.e. α is an algebraic number of degree 2) Since $\alpha^2 = 4\alpha - 1$. We have $f(\alpha) = \alpha^3 + a\alpha^2 + b\alpha + c = (15 + 4 + b)\alpha + (c - a - 4)$. Now, suppose $f(\alpha) \neq 0$, then we have $(15 + 4a + b)(c - a - 4) \neq 0$ since $2 + \sqrt{3}$ is not rational (this is an easy application of rational root theorem). Write $p = |15 + 4a + b|, q = |c - a - 4|$. Now $|\beta f(\alpha)| = |p + q\beta|$ since $\alpha\beta = 1$. By Liouville, we have:

$$\frac{|\beta||f(\alpha)|}{q} = \left| \beta + \frac{p}{q} \right| \geq \frac{c(\beta)}{q^2}.$$

Now, $c(\beta) = \min\{M, 1/3M\} = \min\{\alpha, 1/(3\alpha)\} = 1/(3\alpha)$ since $3\alpha^2 > 1$.

This gives:

$$|f(\alpha)| \geq \frac{1}{\alpha\beta 3q} = \frac{1}{3q}.$$

Finally, we have $3q \leq 3|c - a - 4| \leq 3(24) = 72 < 100$. Therefore, $1/3q > 10^{-2} > 10^{-4}$. This is a contradiction and we are done. $\square$

Another application of Liouville's theorem is to prove that some number is transcendental. To my knowledge, this is the first known transcendental number.

Example 4.29. The number $\alpha = \sum_{n=0}^{\infty} \frac{1}{10^{n!}}$ is transcendental.

Proof. Suppose not, let

$$\sum_{n=0}^k \frac{1}{10^{n!}} = \frac{p_k}{q_k},$$

where $q_k = 10^{k!}$. This implies

$$\left| \alpha - \frac{p_k}{q_k} \right| = \left| \sum_{n=k+1}^{\infty} \frac{1}{10^{n!}} \right| \leq \frac{1}{10^{(k+1)!}} \left(1 + \frac{1}{10^2} + \frac{1}{10^3} + \dots \right),$$

write $S = 1 + 1/10^2 + \dots$, which we can calculate by standard technique in summing geometric series. This implies:

$$\left| \alpha - \frac{p_k}{q_k} \right| \leq \frac{S}{10^{(k+1)!}} = \frac{S}{q_k^{k+1}}.$$

Now, if α were algebraic of degree n , then by Liouville's theorem, there exists a constant $C(\alpha)$ such that

$$\left| \alpha - \frac{p_k}{q_k} \right| \geq \frac{C(\alpha)}{q_k^n}.$$

This implies $S/q_k^{k+1} \geq C(\alpha)/q_k^n$. However, this inequality clearly fails to hold for sufficiently large k . This is a contradiction and hence α is transcendental. $\square$

The following theorem is a far reaching generalization of Liouville's theorem and for which Roth was awarded the Fields Medal:

Theorem 4.18 (Roth). *Let α be an algebraic number of degree $d \geq 2$. Then for any $\epsilon > 0$, then there exists a constant $C(\alpha, \epsilon)$ such that for all $p/q \in \mathbb{Q}$,*

$$\left| \alpha - \frac{p}{q} \right| \geq \frac{C(\alpha, \epsilon)}{q^{2+\epsilon}}.$$

Theorem 4.19 (Roth). *(Variant) Let α be an algebraic number of degree $d \geq 2$. Then for any $\epsilon > 0$ and any constant C , there exists only finitely many $p/q \in \mathbb{Q}$ such that:*

$$\left| \alpha - \frac{p}{q} \right| \leq \frac{C}{q^{2+\epsilon}}.$$

Theorem 4.20. *Fix $m \in \mathbb{Z}$ and a cube-free integer d , then there are only finitely many integer solutions to*

$$x^3 - dy^3 = m.$$

Proof. Let $\alpha = \sqrt[3]{d}$. Then we have $x^3 - dy^3 = (x - \alpha y)(x^2 - \alpha xy + \alpha^2 y^2)$. Note that $x^2 - xy\alpha + \alpha^2 y^2 = (x - \alpha y/2)^2 + (3/4)\alpha^2 y^2 \geq (3/4)\alpha^2 y^2$.

Hence we have:

$$|m| = |x - \alpha y| |x^2 - \alpha xy + \alpha^2 y^2| \geq |x - \alpha y| \frac{3\alpha^2 y^2}{4}.$$

Dividing both sides by $3\alpha^2|y|^3/4$ gives:

$$\left| \frac{x}{y} - \alpha \right| \leq \frac{C}{y^3}, \tag{4.1}$$

where $C = (4m)/(3\alpha^2)$. Now choose $\epsilon > 0$ such that $2 + \epsilon < 3$, then by Roth's theorem, there are only finitely many pairs (x, y) such that the inequality is satisfied, and hence the diophantine equation $x^3 - dy^3 = m$ has only finitely many solutions. $\square$

Theorem 4.21. *Let $n \geq 3$ be an integer, then for any non-zero $a, b, c \in \mathbb{Z}$, the equation:*

$$ax^n + by^n = c$$

has finitely many integer solutions.

Proof. Exercise. $\square$

Theorem 4.22. *If $f(x) \in \mathbb{Z}[x]$ with positive leading coefficient, and with degree $d \geq 2$. Let*

$$S = \sum_{n=1}^{\infty} 1/m^{f(n)}.$$

Then S is irrational for integer $m \geq 2$.

Proof. Suppose $S = a/b$ is rational. Since $f'(x) \rightarrow \infty$ as $x \rightarrow \infty$, there exists an integer t such that $f'(x) > 1$ for all $x > t$. Fix such an integer $t > 2$. Write $a/b = (ta)/(tb) = p/q$, where $q = tb$. Now $S = p/q = S_1 + S_2$, where

$$S_1 = \sum_{n=1}^q 1/m^{f(n)}, S_2 = \sum_{n=q+1}^{\infty} 1/m^{f(n)}.$$

Now consider $qm^{f(q)}S = qm^{f(q)}S_1 + qm^{f(q)}S_2$. Since $qm^{f(q)}S_1 \in \mathbb{Z}$, we must have $qm^{f(q)}S_2 \in \mathbb{Z}$.

Claim 1: Suppose $q > 2$ is an integer such that $f'(x) > 1$ for all $x > q$ then

$$m^{f(q+1)-f(q)} > 2q$$

for any integer $m \geq 2$.

Proof. By the Mean Value Theorem, there exists $c \in (q, q+1)$ such that $f(q+1) - f(q) = qf'(c) > q$ by our choice of q . Now $m^q \geq 2^q > 2q$ for $q > 2$. $\square$

Claim 2: $S_2 < 2/m^{f(q+1)}$

Proof. $S_2 = 1/m^{f(q+1)}A$, where A is bounded by the geometric series with ratio $1/m$. [Note that by our choice of q , for integers $x, y \in (q, \infty)$ with $x > y$, then $f(x) - f(y) \geq 1$.] $\square$

Finally, $qm^{f(q)}S_2 < 2q/m^{f(q+1)-f(q)} < 1$ by the two claims. Therefore, $0 < qm^{f(q)}S_2 < 1$, and hence cannot be an integer. We have a contradiction and hence S is irrational. Done. $\square$

4.12 Some Advanced Theorems

This section samples a collection of advanced theorems that are important and interesting in itself.

Theorem 4.23 (Fermat's Last Theorem). *Suppose $n \geq 3$. If x, y, z are integers and $x^n + y^n = z^n$, then $xyz = 0$.*

Theorem 4.24 (Legendre). *Let a, b, c be non-zero integers such that abc is square-free. Then $ax^2 + by^2 + cz^2 = 0$ have a solution in integers x, y, z not all zero if and only if a, b, c do not have the same sign, and that $-bc, -ac, -ab$ are quadratic residues mod a, b, c respectively.*

Theorem 4.25 (Dirichlet). *Suppose $(a, b) = 1$, then $an + b$ is prime for infinitely many n .*

Example 4.30. Prove that there are infinitely many primitive pythagorean triples (x, y, z) such that z is prime.

Proof. By the primitive pythagorean triple theorem, (x, y, z) is of the form: $(m^2 - n^2, 2mn, m^2 + n^2)$. By Fermat's two squares theorem, any prime p of the form $4k + 1$ is a sum of two squares. Finally, by Dirichlet's theorem, there are infinitely many primes of the form $4k + 1$. Hence there are infinitely many pairs (m, n) such that $m^2 + n^2$ is a prime. We are done. $\square$

Theorem 4.26 (Skolem). *If $d \in \mathbb{Z}, d \neq 0$. Then there exists at most one pair $(x, y) \in \mathbb{Z}^2, y \neq 0$ such that $x^3 + dy^3 = 1$.*

Theorem 4.27 (Catalan). *The equation $a^b - c^d = 1$ has a unique solution in positive integers with $a, b, c, d > 1$, viz. $(a, b, c, d) = (3, 2, 2, 3)$.*

Theorem 4.28 (Fibonacci Power). *1, 8, and 144 are the only perfect powers in the Fibonacci sequence.*

4.13 Problems

Problem 4.1. Solve $11x + 13y = 26$ in integers.

Problem 4.2. Solve $10x - 7y = 17$ in integers.

Problem 4.3. Solve $2x + 3y = 2006$ in positive integers.

Problem 4.4. Let a, b be positive integers such that $(a, b) = 1$. Let A denote the set of all integers expressible as $ax + by$ where x, y are non-negative integers. Show that $c = ab - a - b$ is not in A , and every integer greater than c is in A .

Problem 4.5. Find all n such that $x^2 - y^2 = n$ is solvable in integers.

Problem 4.6. Prove that $x^2 + y^2 = z^4$ has infinitely many integer solutions with $(x, y, z) = 1$.

Problem 4.7. Determine all integer solutions to $x^2 + y^2 = 2z^2$ with $(x, y, z) = 1$.

Problem 4.8. Prove that no Pythagorean Triples belongs to an isosceles right triangle.

Problem 4.9. Solve $x^4 - 2y^2 = 1$ in integers.

Problem 4.10. Show that there does not exist a right triangle with rational sides whose area is 1.

Problem 4.11. Prove theorem 4.2.

Problem 4.12. Prove that the product of three consecutive positive integers is never a square.

Problem 4.13. Prove that the product of four consecutive positive integers is never a square.

Problem 4.14. Prove that there do not exist positive integers x, y such that both $x^2 + y$ and $y^2 + x$ are perfect squares.

Problem 4.15. Let N be a 21-digit number formed by permuting the numbers $1, 2, \dots, 15$. Can N ever be a perfect square?

Problem 4.16. Find the maximal possible integer x such that $4^{27} + 4^{1000} + 4^x$ is a perfect square.

Problem 4.17. Let x, y be positive integers and suppose $2xy \mid x^2 + y^2 - x$. Prove that x is a square.

Problem 4.18. Find all positive integer solutions to $10^x - 9^y = 1$.

Problem 4.19. Prove that $3a^2 - 8ab + 9b^2 = 11006$ is not solvable in integers.

Problem 4.20. Prove that $x^3 - 3xy^2 + y^3 = 2891$ is not solvable in integers.

Problem 4.21. Prove that $x^2 + y^2 - 19xy = 19$ is not solvable in integers.

Problem 4.22. Solve $x^2 - 4xy + 5y^2 = 169$ in integers.

Problem 4.23. Solve $y^2 + 3x^2y^2 = 20x^2 + 517$ in positive integers.

Problem 4.24. Prove that $y^2 = x^5 - 4$ has no integer solutions.

Problem 4.25. Show that $1998^2x^2 + 1997x + 1995 - 1998x^{1998} = 1998y^4 + 1993y^3 - 1991y^{1998} - 2001y$ has no solution in integers. [Hint: 1997 is a prime.]

Problem 4.26 (IMO 1994). Determine all ordered pairs (m, n) of positive integers such that

$$\frac{n^3 + 1}{mn - 1}$$

is an integer.

Problem 4.27 (IMO 2003). Determine all pairs of integers (a, b) such that

$$\frac{a^2}{2a^2b - b^3 + 1}$$

is a positive integer.

Problem 4.28 (IMO 1992). Find all positive integers a, b and c such that $1 < a < b < c$ and $(a-1)(b-1)(c-1) \mid abc-1$.

Problem 4.29. Solve $x^n + (x+1)^n = (x+2)^n$ in positive integers

Problem 4.30. Find all natural triples (x, y, z) such that $x^4 - y^4 = z^2$.

Problem 4.31. Show that $x^2 + y^2 = 3$ is not solvable in rational numbers x, y .

Problem 4.32. Show that $x^2 + y^2 + z^2 = x^2y^2$ has no integer solution other than $x = y = z = 0$.

Problem 4.33. Solve $3x^2 + 1 = 4y^3$ for integers.

Problem 4.34. Prove that $5x^3 + 11y^3 + 13z^3 = 0$ has no solution in integers other than $x = y = z = 0$.

Problem 4.35. Given $a_0 = 0, a_1 = 2, a_{n+2} = 4a_{n+1} + a_n$ for $n \geq 0$. Find all n such that a_n is a perfect square.

Problem 4.36. Let $a_1 = 1, a_{n+1} = 5a_n^3 - 3a_n$. Find all n such that a_n is a perfect square.

Problem 4.37. Suppose m, n are odd positive integers and $m^2 - n^2 + 1 \mid n^2 - 1$. Prove that $m^2 - n^2 + 1$ is a perfect square.

Problem 4.38. Suppose x, y are positive integers such that $xy \mid x^2 + y^2 + 1$, prove that $(x^2 + y^2 + 1)/xy = 3$.

Problem 4.39. Find all natural triples (x, y, z) such that $x^3 + y^3 + z^3 = x + y + z = 3$. What if x, y, z are positive reals? What if x, y, z are integers?

Problem 4.40. Find all positive integer k such that: $a^2 + b^2 = kab$ is solvable in positive integers.

Problem 4.41. Suppose k is a positive integer such that $a^2 + b^2 + c^2 + d^2 = kabcd$ is solvable in positive integers. Prove that $k \leq 4$.

Problem 4.42. Solve $12^x - 5^y = 7$ in positive integers.

Problem 4.43. Solve $3^x + 4^y = 5^z$ in positive integers.

Problem 4.44. Solve $x^2 + y^2 - 5xy - 5 = 0$ in positive integers.

Problem 4.45. Solve $2^x + 3^y = 5^z$ in positive integers.

Problem 4.46. Solve $a^3 + b^3 + 3ab = 1$ in integers.

Problem 4.47. Find all pairs (x, y) of positive integers that satisfy: $y^2 = x^3 + 16$.

Problem 4.48 (Putnam 2001). Prove that there are unique positive integers a, n such that $a^{n+1} - (a+1)^n = 2001$.

Problem 4.49. Prove theorem 4.21.

Problem 4.50. Prove that $\sum_{n=1}^{\infty} 2^{-3^n}$ is transcendental.

Problem 4.51. Prove that $\sum_{n=1}^{\infty} 2^{-f(n)}$ is transcendental whenever

$$\lim_{n \rightarrow \infty} \frac{f(n+1)}{f(n)} > 2.$$

Chapter 5

Gaussian Integers, and the Two Square Theorem

5.1 Arithmetic in $\mathbb{Z}[i]$

Definition 5.1 (Gaussian Integer). *The Gaussian Integers is defined to be $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$.*

Example 5.1. $1, 2, 3, 2 + i, 4 + 5i$ are all Gaussian integers, while $1/2 + i$ is NOT a Gaussian integer because $1/2 \notin \mathbb{Z}$.

As it turns out, the arithmetic on $\mathbb{Z}[i]$ is very similar to that on $\mathbb{Z}$. We will next define the analogous terms as in the case of $\mathbb{Z}$.

Definition 5.2 (Unit). *An element $\alpha = a + bi \in \mathbb{Z}[i]$ is a unit if there is another element $\beta = c + di \in \mathbb{Z}[i]$ such that $\alpha\beta = 1$.*

It is easy to see that $1, -1, i, -i$ are units in $\mathbb{Z}[i]$, the question is: are there more? The answer is no and we shall see why shortly.

Definition 5.3 (Prime). *An element $\alpha = a + bi \in \mathbb{Z}[i]$ is a prime if it can be written as $\alpha = \beta\gamma$ where β, γ are non-unit element in $\mathbb{Z}[i]$. If α is a prime and it is actually in $\mathbb{Z}$, we called these **rational primes**.*

While it is tempting to assume that primes in $\mathbb{Z}$ are automatically primes in $\mathbb{Z}[i]$. It is not the case. For example, we see that $2 = -i(1+i)^2$ is a prime in $\mathbb{Z}$, but not a prime in $\mathbb{Z}[i]$. [Technically, we need to show that $1 + i$ is not a unit, but let's assume it for the moment.]

Definition 5.4 (Norm). *The norm of $\alpha = a + bi$ is defined as $N(\alpha) = (a + bi)(a - bi) = \alpha\bar{\alpha}$. Note that it is a straight forward calculation that $N(\alpha\beta) = N(\alpha)N(\beta)$ for all $\alpha, \beta \in \mathbb{Z}[i]$.*

With the help of norm, we first determine the units in $\mathbb{Z}[i]$.

Theorem 5.5. $\alpha \in \mathbb{Z}[i]$ is a unit iff $N(\alpha) = 1$.

Proof. Suppose $\alpha = a + bi$ is a unit. Then there exists $\beta \in \mathbb{Z}[i]$ such that $\alpha\beta = 1$. Taking norm on both sides give $N(\alpha)N(\beta) = N(1) = 1$. Notice that both $N(\alpha), N(\beta)$ are positive integers. This says $N(\alpha) = N(\beta) = 1$.

Conversely, suppose $N(\alpha) = 1$. Write $\alpha = a + bi$. We have $1 = N(\alpha) = \alpha\bar{\alpha} = (a + bi)(a - bi)$. This says $\alpha = a + bi$ is a unit. ($\beta = \bar{\alpha}$ here) $\square$

Corollary 5.6 (Units in $\mathbb{Z}[i]$). *The units in $\mathbb{Z}[i]$ are $\{\pm 1, \pm i\}$.*

Proof. This amounts to solve $a^2 + b^2 = 1$ for integers a, b , which we will leave to the reader to verify that $(a, b) = (1, 0), (0, 1), (-1, 0), (0, -1)$ are the only solutions. $\square$

Lemma 5.7. *If $\alpha \in \mathbb{Z}[i]$, and $N(\alpha) = p$ is a rational prime, then α is prime in $\mathbb{Z}[i]$.*

Proof. Suppose not, then $\alpha = \beta\gamma \in \mathbb{Z}[i]$ with $N(\beta), N(\gamma) > 1$. Now $p = N(\alpha) = N(\beta)N(\gamma)$. Note that both $N(\beta), N(\gamma)$ are both positive integers > 1 whose product is a rational prime p . This is clearly impossible. $\square$

Remark: All theorems and properties concerning divisibility of $\mathbb{Z}$ carries over to $\mathbb{Z}[i]$, and we will be using these properties freely here.

Remark: The converse of this lemma is far from being true, as the next theorem suggests

Theorem 5.8 (Primes in $\mathbb{Z}[i]$). *Up to multiplication by a unit, the primes in $\mathbb{Z}[i]$ falls in exactly one of the following three categories.*

1. $1 + i$.
2. Rational primes of the form $q = 4k + 3$, for some positive integer k .
3. $\pi \in \mathbb{Z}[i]$ such that $N(\pi) = p$ is a rational prime of the form $p = 4k + 1$ for some positive integer k .

Proof. It is trivial to verify that $N(1 + i) = 2$ and (1) follows. Let q be a rational prime that is $3 \pmod{4}$. If $q = \alpha\beta$ for some non-unit $\alpha, \beta \in \mathbb{Z}[i]$. Then $q^2 = N(q) = N(\alpha\beta) = N(\alpha)N(\beta)$. This says $N(\alpha) = N(\beta) = q$. But we already know that sum of two squares are never $3 \pmod{4}$ and hence (2) follows. Finally, suppose $\pi \in \mathbb{Z}[i]$ with $N(\pi) = p \equiv 1 \pmod{4}$. It is clear that such elements are prime, what we need to show is that rational prime of the form $4k + 1$ is not prime in $\mathbb{Z}[i]$. By Corollary 3.18, $p \mid n^2 + 1$ for some n . Thus $p \mid (n + i)(n - i)$. Since $\pi \mid p$, we have $\pi \mid n + i$ or $\pi \mid n - i$. If p were prime in $\mathbb{Z}[i]$, then $p \mid (n + i)(n - i)$ implies $p \mid n + i$ or $p \mid n - i$, but this is not the case since $(n + i)/p$ is in $\mathbb{Z}[i]$ iff both n/p and $1/p$ are integers. Hence $p = \pi\bar{\pi}$. Hence (3) follows. $\square$

Theorem 5.9 (Unique Factorization in $\mathbb{Z}[i]$). *Every $\alpha \in \mathbb{Z}[i]$ can be uniquely factorize as primes up to units.*

Theorem 5.10 (Fermat's Two Square Theorem). *A prime p is expressible as a sum of two squares iff $p \equiv 1 \pmod{4}$.*

Proof. This is a direct consequence of Theorem 5.8 part (3). □

Example 5.2. Solve $y^2 + 1 = x^3$ for integers.

Proof. First note that y is even, by considering mod 4. We then work in $\mathbb{Z}[i]$, factorize the equation as $(y+i)(y-i) = x^3$. First show $d = (y+i, y-i)$ is a unit. Note that $d \mid y+i, y-i$ says $d \mid 2i$. Since y is even, we also have $d \mid y$ and this implies $d \mid i$. Hence d is a unit. Therefore each of $y+i$ and $y-i$ is a cube in $\mathbb{Z}[i]$. Writing $y+i = u(a+bi)^3$, where u is a unit. Since the units in $\mathbb{Z}[i]$ are precisely $\pm 1, \pm i$ and they are all cubes, we may WOLOG assume $u = 1$. Expanding the right hand side gives $y+i = a^3 + 3abi + 3ab^2 - b^3i$. Comparing the imaginary part gives: $1 = 3a^2b - b^3 = b(3a^2 - b^2)$. This implies $b = \pm 1$, from here it is easy to see that the equation has no solutions. □

Example 5.3. As another application of Gaussian integers, we will give another proof of the primitive Pythagorean Triples.

Proof. Write $x^2 + y^2 = (x+iy)(x-iy)$. Let $d = (x+iy, x-iy)$. Hence $d \mid 2x$. By simple congruence argument, we know that z must be odd, and one of x, y must be odd. Let us WOLOG assume x is odd. Thus $d \mid 2x$ and $d \mid z$ implies d must be a unit. [Notice that the assumption $(x, y, z) = 1$ pass on even to the Gaussian Integers. (Convince yourself that this is indeed the case!)] This implies $x+iy = (m+in)^2$ and $x-iy = (m-in)^2$. Expanding and comparing real and imaginary parts give:

$$x = m^2 - n^2, \quad y = 2mn.$$

The condition that m and n have opposite signs is satisfied because x is odd. Done. □

5.2 Problems

Problem 5.1. a) Show that the product of sum of two squares is again a sum of two squares. Equivalently, prove that

$$(a^2 + b^2)(c^2 + d^2) = (x^2 + y^2),$$

where x, y are in terms of a, b, c and d .

b) Prove the following version of Fermat's Two Squares Theorem: An Integer n is expressible as a sum of two squares if and only if the primes factors $q \equiv 3 \pmod{4}$ of n has even exponent.

Problem 5.2. Solve $y^2 + 1 = x^5$ for integers.

Problem 5.3. Solve $y^2 + 4 = x^3$ for integers.

Problem 5.4. Given $a, b, x, y \in \mathbb{N}$ such that $a^2 + b^2 \mid ax + by$. Prove that $(x^2 + y^2, a^2 + b^2) > 1$.

Problem 5.5. Prove that there is at most one way to write a prime p into the sum of two positive integer squares. (Here we assume the representation $1^2 + 2^2 = 2^2 + 1^2 = 5$ as the same.)

Chapter 6

Quadratic Reciprocity

Reciprocity Laws have fascinated a lot of mathematicians, including Legendre, Euler, Gauss, Jacobi, Eisenstein, Hilbert, Kummer, Dedekind, Kronecker, Hasse and Artin. But what motivates the study of reciprocity law? Basically, mathematicians want to study congruence equations or high order. The first non-trivial case is the quadratic case. (i.e., $x^2 \equiv a \pmod{p}$, for p prime.) Naturally, there are two questions to ask concerning the solutions of the congruence equation. Are there any solutions? If so, what are the solutions? There is one simple way to answer both questions, since we are only working in modulo p , we may simply try all the congruence classes modulo p (there are only finitely many) to see if the equation is solvable. Of course, such brute-force method was not appealing to mathematicians and we hence sought more elegant and efficient method to answer our questions. To this end, Legendre introduced a symbol named after him to determine whether such equation is solvable. Now, to answer the existence of solution, we have reduced the problem to calculation of the Legendre Symbol. Mathematicians sought a way to efficiently calculate the Legendre Symbol, and here comes the birth of:

Theorem (Quadratic Reciprocity Law). *Let $p, q \in \mathbb{N}$ be distinct odd primes; then:*

$$\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2} \frac{q-1}{2}} \left(\frac{q}{p}\right).$$

Moreover, we have:

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}} \quad \left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}};$$

these are called the first and second supplementary law, respectively.

It was Euler who conjectured this theorem, and Legendre, who attempted to prove it, but his proof was incomplete. The first complete proof of quadratic reciprocity was by

Gauss on April 8, 1796. Gauss himself gave 8 different proofs to it, in which 6 of them were published. So, the quadratic case was settled, and we naturally move to higher degree congruence equations and hoping for higher reciprocity laws. While searching for Higher Reciprocity Laws, it marks the development of Algebraic Number Theory, for the study of reciprocity laws go beyond rational numbers, to algebraic integers during the process of generalizing to higher degree.

Definition 6.1. An integer a , where $(a, m) = 1$, is called a **quadratic residue** $(\text{mod } m)$ if $x^2 \equiv a \pmod{m}$ has a solution. If the congruence has no solution, then a is called a **quadratic non-residue** $(\text{mod } m)$.

Definition 6.2. Let p be a prime. The **Legendre symbol** $\left(\frac{a}{p}\right)$ is defined to be:

$$\left(\frac{a}{p}\right) = \begin{cases} +1 & \text{if } a \text{ is a quadratic residue} \\ -1 & \text{if } a \text{ is a quadratic non-residue} \\ 0 & \text{if } p \mid a \end{cases}$$

Proposition 6.3 (Properties of Legendre symbol). Suppose p is an odd prime, and $a, b \in \mathbb{Z}$. Then:

1. $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right).$
2. $a \equiv b \pmod{p} \Rightarrow \left(\frac{a}{p}\right) = \left(\frac{b}{p}\right).$
3. If $(a, p) = 1$ then $\left(\frac{a^2}{p}\right) = 1, \left(\frac{a^2b}{p}\right) = \left(\frac{b}{p}\right).$

Proof. These are simple consequences of the definition. □

Corollary 6.4. Let $f(x) \in \mathbb{Z}[z]$. Let p be a prime. Let $a, b \in \mathbb{Z}$ such that $(a, p) = 1$. Then:

$$\sum_{k=0}^{p-1} \left(\frac{f(ak+b)}{p}\right) = \sum_{k=0}^{p-1} \left(\frac{f(k)}{p}\right).$$

Proof. Since legendre symbol only depends on congruence classes mod p . It suffices to show that $ak + b \pmod{p}$ runs through 0 to $p-1 \pmod{p}$ as k runs through 0 to $p-1$. First of all, it is obvious that $\{ak + b \pmod{p} \mid 0 \leq k \leq p-1\} = \{ak \pmod{p} \mid 0 \leq k \leq p-1\}$. Next, since $(a, p) = 1$, we have $\{ak \pmod{p} \mid 0 \leq k \leq p-1\} = \{k \pmod{p} \mid 0 \leq k \leq p-1\}$. We are done.

We can also give a shorter proof by saying that it is equivalent to show that $\varphi : \mathbb{F}_p \rightarrow \mathbb{F}_p$, where $\varphi(k) = ak + b$ is an isomorphism, which is clear because $\psi(k) = (k - b)/a$ is an inverse. $\square$

Proposition 6.5 (Euler's Criterion). *Suppose p is a prime, a an integer such that $(a, p) = 1$. Then we have:*

$$a^{\frac{p-1}{2}} \equiv \begin{cases} +1 \pmod{p} & \text{if } a \text{ is a quadratic residue} \pmod{p} \\ -1 \pmod{p} & \text{if } a \text{ is a quadratic non-residue} \pmod{p} \end{cases}$$

Proof. Suppose $\left(\frac{a}{p}\right) = 1$. Then $x^2 \equiv a \pmod{p}$ has a solution, call it x_0 . Then by Fermat's

Little Theorem, we have: $a^{\frac{p-1}{2}} \equiv x_0^{p-1} \equiv 1 \pmod{p}$. Now suppose $\left(\frac{a}{p}\right) = -1$, then $x^2 \equiv a \pmod{p}$ had no solution. For each j , choose j' , where $1 \leq j, j' < p$, such that $jj' \equiv a \pmod{p}$. We pair j with j' . We note that $j \not\equiv j' \pmod{p}$ for all j . For if they were equal, that means $x^2 \equiv a \pmod{p}$ has a solution, (i.e., j). Now if we multiply all the j and their corresponding j' , we have $(p-1)!$ on one hand, and $a^{\frac{p-1}{2}} \pmod{p}$ on the other hand since we have $(p-1)/2$ pairs of j and j' . Equating we get: $a^{\frac{p-1}{2}} \equiv (p-1)! \pmod{p}$. The assertion now follows from Wilson's Theorem, which asserts that $(p-1)! \equiv -1 \pmod{p}$ if and only if p is a prime. $\square$

Corollary 6.6. *There are as many quadratic residues as quadratic non-residues.*

Proof. By Euler's criterion, each solution to $a^{(p-1)/2} \equiv 1 \pmod{p}$ is a quadratic residue. Since the equation has $(p-1)/2$ solutions, there are $(p-1)/2$ quadratic residues and therefore there are $(p-1) - (p-1)/2 = (p-1)/2$ quadratic non-residues. $\square$

Corollary 6.7. *For an odd prime p , we have*

$$\sum_{k=1}^{p-1} \left(\frac{k}{p}\right) = 0.$$

Proof. Since there are as many residues as non-residues. The result follows immediately. $\square$

Proposition 6.8. *Let p be an odd prime. Then we have:*

$$\sum_{k=1}^{p-2} \left(\frac{k(k+1)}{p} \right) = -1.$$

Proof.

$$\sum_{k=1}^{p-2} \left(\frac{k(k+1)}{p} \right) = \sum_{k=1}^{p-2} \left(\frac{k^2(1+k^{-1})}{p} \right) = \sum_{k=1}^{p-2} \left(\frac{1+k^{-1}}{p} \right).$$

Since $k \not\equiv -1 \pmod{p}$, we have $1+k^{-1} \not\equiv 0 \pmod{p}$. Also, we have $1+k^{-1} \not\equiv 1 \pmod{p}$. Thus $1+k^{-1} \pmod{p}$ runs through 2 to $p-1 \pmod{p}$ as k runs through 1 to $p-2$. Hence:

$$\sum_{k=1}^{p-2} \left(\frac{1+k^{-1}}{p} \right) = \sum_{a=2}^{p-1} \left(\frac{a}{p} \right) = - \left(\frac{1}{p} \right) = -1.$$

□

Theorem 6.9 (Law of Quadratic Reciprocity). *Suppose p, q are distinct odd primes. Then we have:*

$$\left(\frac{p}{q} \right) = (-1)^{\frac{p-1}{2} \frac{q-1}{2}} \left(\frac{q}{p} \right).$$

Equivalently, we have:

$$\left(\frac{p}{q} \right) = \begin{cases} \left(\frac{q}{p} \right) & \text{if } p \equiv 1 \pmod{4}, \text{ or } q \equiv 1 \pmod{4} \\ - \left(\frac{q}{p} \right) & \text{if } p \equiv q \equiv 3 \pmod{4} \end{cases}$$

Theorem 6.10 (Supplementary Laws). *Let p be an odd prime. Then we have:*

$$\left(\frac{-1}{p} \right) = (-1)^{\frac{p-1}{2}} \quad \left(\frac{2}{p} \right) = (-1)^{\frac{p^2-1}{8}}.$$

Equivalently, we have:

$$\left(\frac{-1}{p} \right) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4} \\ -1 & \text{if } p \equiv 3 \pmod{4} \end{cases}$$

$$\left(\frac{2}{p} \right) = \begin{cases} 1 & \text{if } p \equiv \pm 1 \pmod{8} \\ -1 & \text{if } p \equiv \pm 3 \pmod{8} \end{cases}$$

Lemma 6.11. *Let p be an odd prime and $a \in \mathbb{Z}$ such that $(a, p) = 1$. Then the total number of solutions to $ax^2 + bx + c \equiv 0 \pmod{p}$ equals:*

$$1 + \left(\frac{b^2 - 4ac}{p} \right).$$

Proof. By completing square, it is clear that $ax^2 + bx + c \equiv 0 \pmod{p}$ is solvable iff $b^2 - 4ac$ is a quadratic residue mod p . The result now follows. $\square$

Corollary 6.12. *Let p be an odd prime. Then the total number of solutions to $x^2 - y^2 \equiv a \pmod{p}$ equals:*

$$\sum_{y=0}^{p-1} 1 + \left(\frac{y^2 + a}{p} \right).$$

Proof. This follows immediately from corollary 6.11. $\square$

Lemma 6.13. *Let p be an odd prime. Then the total number of solutions to $x^2 - y^2 \equiv a \pmod{p}$ equals:*

$$\begin{cases} p - 1 & \text{if } p \nmid a \\ 2p - 1 & \text{if } p \mid a. \end{cases}$$

Proof. Let $u = x + y, v = x - y$. Then the equation can be rewritten as $uv \equiv a \pmod{p}$. Now suppose $p \nmid a$. Then for each nonzero v , there is a unique solution u . Hence there are exactly $p - 1$ solutions because there are exactly $p - 1$ choices for v .

Now suppose $p \mid a$, then the equation is now $uv \equiv 0 \pmod{p}$. Therefore, the number of solutions equals: $\#\{(u, 0) \mid u \in \mathbb{F}_p\} + \#\{(0, v) \mid v \in \mathbb{F}_p\} - \#\{(0, 0)\} = 2p - 1$. $\square$

Corollary 6.14. *Let p be an odd prime. Then we have:*

$$\sum_{y=0}^{p-1} \left(\frac{y^2 + a}{p} \right) = \begin{cases} -1 & \text{if } p \nmid a \\ p - 1 & \text{if } p \mid a. \end{cases}$$

Proof. This follows immediate from corollary 6.12, and lemma 6.13. $\square$

Example 6.1. Let $p \equiv 1 \pmod{4}$ be a prime. Prove that

$$\sum_{k=0}^{p-1} k \left(\frac{k}{p} \right).$$

Proof. Let S be the desired sum. Then we have:

$$S = \sum_{k=0}^{p-1} (p-k) \left(\frac{p-k}{p} \right) = \sum_{k=0}^{p-1} (p-k) \left(\frac{-k}{p} \right) = \sum_{k=0}^{p-1} (p-k) \left(\frac{k}{p} \right),$$

where the last equality holds since $(-1/p) = 1$ because $p \equiv 1 \pmod{4}$. Therefore, we have:

$$S = p \sum_{k=0}^{p-1} \left(\frac{k}{p} \right) - S = -S,$$

by corollary 6.7. This implies $2S = 0$, and hence $S = 0$ as claimed. $\square$

6.1 Jacobi Symbol and Generalized Quadratic Reciprocity

Definition 6.15. Let $Q = q_1 q_2 \dots q_s$, where q_i are odd primes (not necessarily distinct). We define the **Jacobi Symbol**:

$$\left(\frac{P}{Q} \right) = \prod_{i=1}^s \left(\frac{P}{q_i} \right).$$

Remark: If Q is prime, then the Jacobi Symbol and Legendre Symbol are the same. If P is quadratic residue modulo Q , then $\left(\frac{P}{q_j} \right) = 1$ for each j , and hence $\left(\frac{P}{Q} \right) = 1$. However, $\left(\frac{P}{Q} \right) = 1$ does not imply P is a quadratic residue of Q . You also need each $\left(\frac{P}{q_j} \right) = 1$ for all j . For example, $\left(\frac{2}{15} \right) = 1$, but $x^2 \equiv 2 \pmod{15}$ has no solution.

Jacobi symbol satisfies the obvious properties enjoyed by the Legendre Symbol. We have:

Proposition 6.16 (Properties of Jacobi Symbol). *Suppose Q , and Q' are odd. Then:*

1. $\left(\frac{P}{Q} \right) \left(\frac{P}{Q'} \right) = \left(\frac{P}{QQ'} \right).$
2. $\left(\frac{P}{Q} \right) \left(\frac{P'}{Q} \right) = \left(\frac{PP'}{Q} \right).$

$$3. \text{ If } (P, Q) = 1, \text{ then } \left(\frac{P^2}{Q}\right) = \left(\frac{P}{Q^2}\right) = 1.$$

$$4. \text{ If } (PP', QQ') = 1, \text{ then } \left(\frac{P'P^2}{Q'Q^2}\right) = \left(\frac{P'}{Q'}\right).$$

$$5. P' \equiv P \pmod{Q} \Rightarrow \left(\frac{P'}{Q}\right) = \left(\frac{P}{Q}\right).$$

Proof. These are all direct consequences of the definition of Jacobi symbol and properties of Legendre Symbol. $\square$

Theorem 6.17. *If Q is odd and $Q > 0$, then:*

$$\left(\frac{-1}{Q}\right) = (-1)^{\frac{Q-1}{2}} \quad \text{and} \quad \left(\frac{2}{Q}\right) = (-1)^{\frac{Q^2-1}{8}}.$$

Theorem 6.18 (Generalized Quadratic Reciprocity). *If P, Q are odd and positive and if $(P, Q) = 1$, then:*

$$\left(\frac{P}{Q}\right) \left(\frac{Q}{P}\right) = (-1)^{\frac{P-1}{2} \frac{Q-1}{2}},$$

6.2 Problems

Problem 6.1. Which of the following congruence equations have solutions?

(a) $x^2 \equiv 1 \pmod{3}$.

(b) $x^2 + 8x + 18 \equiv -1 \pmod{17}$.

Problem 6.2. Solve $2x^2 + 3x + 1 \equiv 0 \pmod{7}$.

Problem 6.3. Show that the odd prime divisors of $n^2 + 1$ are of the form $12k + 1$ or $12k + 5$.

Problem 6.4. Let p be a prime of the form $6^n + 1$. Find all n such that 5 is a quadratic residue mod p .

Problem 6.5. For any prime $p > 5$. Show that it is always possible to find two consecutive integers which are quadratic residues mod p , as well as two consecutive integers which are quadratic non-residues mod p .

Problem 6.6. Show that $x^2 - 33y^2 = 5$ has no integer solutions.

Problem 6.7. Characterize all prime numbers $p > 11$ such that $x^2 \equiv 11 \pmod{p}$ is solvable.

Problem 6.8. Let $p \equiv 1 \pmod{4}$ be a prime. Show that:

$$\sum_{k=1}^{(p-1)/2} \left(\frac{j}{p} \right) = 0.$$

Problem 6.9. Let p be an odd prime. $a, b, c \in \mathbb{Z}$ such that $(a, p) = 1$ Prove that:

$$\sum_{x=0}^{p-1} \left(\frac{ax^2 + bx + c}{p} \right) = \begin{cases} - \left(\frac{a}{p} \right) & \text{if } p \nmid b^2 - 4ac \\ (p-1) \left(\frac{a}{p} \right) & \text{if } p \mid b^2 - 4ac. \end{cases}$$

Problem 6.10. Let $p \equiv 1 \pmod{4}$ be a prime. Prove that:

$$\sum_{k=0, (k/p)=1}^{p-1} k = \frac{p(p-1)}{4}.$$

Problem 6.11. Let $p \equiv 3 \pmod{4}$ be a prime. Prove that:

$$\sum_{k=1}^{p-1} k^2 \left(\frac{k}{p} \right) = p \sum_{k=1}^{p-1} k \left(\frac{k}{p} \right).$$

Problem 6.12. Let $a, b \in \{1, -1\}$, p be an odd prime, and

$$N(a, b) = \# \left\{ k \mid a \leq k \leq p-2, \left(\frac{k}{p} \right) = a, \left(\frac{k+1}{p} \right) = b \right\}.$$

Prove that

$$N(a, b) = \frac{1}{4} (p-2-b-ab-a(-1)^{(p-1)/2}).$$

Hint: Define

$$a_p(k) = \begin{cases} 1 & \text{if } \left(\frac{k}{p} \right) = a, \left(\frac{k+1}{p} \right) = b \\ 0 & \text{otherwise.} \end{cases}$$

Show that

$$a_p(k) = \frac{1}{4} \left(1 + a \left(\frac{k}{p} \right) \right) \left(1 + b \left(\frac{k+1}{p} \right) \right).$$

Chapter 7

Lattice Points Counting

Proposition 7.1. *Let $x \in \mathbb{R}$. Here are some properties of the greatest integer function $[x]$.*

1. $x = [x] + \{x\}, 0 \leq \{x\} < 1$.
2. $[x + n] = n + [x]$ for all integers n .
3. $[x] + [-x] = -1$ unless $x \in \mathbb{Z}$.
4. *Hermite's Identity*

$$[x] + [x + 1/n] + [x + 2/n] + \dots [x + (n - 1)/n] = [nx].$$

Proof. (1) and (2) are trivial. For (3), write $[-x] = -([x] + \{x\}) = -[x] + [-\{x\}]$. Now, unless $x \in \mathbb{Z}$, we have $0 < \{x\} < 1$, thus $-1 < -\{x\} < 0$. This implies $[-\{x\}] = -1$. Hence $[x] + [-x] = [x] - [x] - 1 = -1$, unless $x \in \mathbb{Z}$. For (4), write $x = [x] + k/n + \alpha$, where $0 \leq k < n$, $0 < \alpha < 1/n$. Then we have $x + i/n = [x] + (k + i)/n + \alpha$. Hence we have: $[x + i/n] = [x] + [(k + i)/n + \alpha]$. Now $i/n \leq (k + i)/n < (n + i)/n = 1 + i/n$. Hence, $i/n \leq (k + i)/n + \alpha < 1 + i/n + 1/n = 1 + (i + 1)/n \leq 2$. Therefore $[(k + i)/n + \alpha] = 0$ unless $i \geq n - k$, in which case $[(k + i)/n + \alpha] = 1$. Therefore the left hand side equals $n[x] + k$, since there are exactly k terms where $i \geq n - k$. Now, the right hand side equals $[nx] = [n[x] + k + n\alpha] = n[x] + k + [n\alpha] = n[x] + k$, since $0 < n\alpha < 1$. Done. $\square$

Theorem 7.2 (Lattice Point Counting). *Let $f(x)$ be a strictly increasing function with inverse $f^{-1}(x)$. Suppose $f(0) = 0, f(a) = b$, and both $a, b > 0$. Let L be the total number of lattice points on the curve $f(x)$ from $O = (0, 0)$, to $P = (a, b)$. Then we have:*

$$\sum_{k=1}^{[a]} [f(k)] + \sum_{k=1}^{[b]} [f^{-1}(k)] - L = [a][b].$$

Example 7.1. Let $[x]$ denote the greatest integer not exceeding x , and $p, q \in \mathbb{N}$ such that $(p, q) = 1$. Prove that:

$$\sum_{x=0}^{q-1} \left[\frac{px}{q} \right] = \frac{(p-1)(q-1)}{2}.$$

Proof. Let $f(x) = px/q$, $O = (0, 0)$, $P = (p-1, q-1)$. Since $(p, q) = 1$ we have $L = 0$. Hence by lattice point counting theorem, we have:

$$\sum_{x=0}^{p-1} \left[\frac{px}{q} \right] + \sum_{x=0}^{q-1} \left[\frac{qx}{p} \right] = (p-1)(q-1).$$

By symmetry, we have the two summand on the left hand side are the same and the assertion follows immediately. $\square$

Example 7.2. Find a formula for:

$$\sum_{k=1}^{n^2} [\sqrt{k}].$$

Proof. Let $f(x) = \sqrt{x}$, $O = (0, 0)$, $P = (n^2, n)$, $L = n$. We now apply lattice point counting theorem to get:

$$\sum_{k=1}^{n^2} [\sqrt{k}] + \sum_{k=1}^n [k^2] - n = [n][n^2].$$

Thus we have

$$\sum_{k=1}^{n^2} [\sqrt{k}] = n^3 + n - \frac{n(n+1)(2n+1)}{6}.$$

$\square$

Example 7.3. Prove that

$$\sum_{k=2}^n [\sqrt[k]{n}] = \sum_{k=2}^n [\log_k n].$$

Proof. Consider the lattice points in the region bounded by $y = n^x$, $x = 2$, $y = 2$. (including the points on the boundary) For each vertical line $x = k$, it intersects the curve $y = n^x$ at $A = (k, \sqrt[k]{n})$. Number of lattice points on this line is $[\sqrt[k]{n}] - 1$. Total number of lattice points in the region equals $\sum_{k=2}^n ([\sqrt[k]{n}] - 1)$. On the other hand, for each horizontal line $y = h$, it intersects the curve $x = \log_n y$ at $B = (\log_h n, h)$. Number of lattice points on this line is $[\log_h n] - 1$. Summing from $h = 2$ to n and equating gives the result at once. $\square$

7.1 Problems

Problem 7.1. Compute

$$\sum_{k=0}^{n^3} [\sqrt[3]{k}].$$

Problem 7.2. Show that for all positive integers m and n ,

$$\gcd(m, n) = m + n - mn + 2 \sum_{k=0}^{m-1} \left\lfloor \frac{kn}{m} \right\rfloor.$$

Problem 7.3. Prove that

$$\frac{1}{2} \sum_{k=1}^n \left\lfloor \frac{n}{k} \right\rfloor \left(\left\lfloor \frac{n}{k} + 1 \right\rfloor \right) = \sum_{k=1}^n k \left\lfloor \frac{n}{k} \right\rfloor.$$

Chapter 8

Lagrange's Four Square Theorem

Theorem 8.1 (Lagrange). *Every positive integer n is a sum of 4 non-negative integers.*

Lemma 8.2. *If p is prime, then $x^2 + y^2 + 1 \equiv 0 \pmod{p}$ is solvable.*

Proof. Let S denote the set of squares mod p , and $S' = \{-1 - x \mid x \in S\}$. It is clear that $|S| = (p+1)/2$. We claim that $|S'| = (p+1)/2$. To see this, it suffices to show that $-1 - x \equiv -1 - y \pmod{p}$ iff $x \equiv y \pmod{p}$. This is clear and thus $S \cap S'$ is non-empty and there exists $x^2 \equiv -1 - y^2 \pmod{p}$. This concludes the proof. $\square$

Corollary 8.3. *If p is a prime, then there is an integer $0 < m < p$ such that $mp = 1 + x^2 + y^2$ has integer solutions.*

Proof. If p is a prime, then the lemma says there exists integer m such that $mp = 1 + x^2 + y^2$ has an integral solution. Furthermore, we may choose our solution x, y in the lemma to satisfy $|x| < p/2, |y| < p/2$. Thus $mp < 1 + p^2/4 + p^2/4 = 1 + p^2/2$. This implies $m < p$. $\square$

Lemma 8.4. *Suppose for a prime p , there is an integer $1 < m < p$ such that mp is the sum of four squares. Then there is an $0 < n < m$ such that np is the sum of four squares.*

Proof. Write $mp = x_1^2 + x_2^2 + x_3^2 + x_4^2$. Let $x_i \equiv y_i \pmod{m}$ with $-m/2 < y_i \leq m/2$. Then we have $y_1^2 + \dots + y_4^2 \equiv 0 \pmod{m}$ and there exists an integer $r \geq 0$ such that $rm = y_1^2 + \dots + y_4^2$. Now $y_i^2 \leq m^2/4$ implies $rm \leq m^2$. Hence $r \leq m$. First note that if $r = 0$, then $y_i = 0$. Since $x_i \equiv y_i \pmod{m}$, this implies $m \mid p$, which is a contradiction for p is a prime and $1 < m < p$. Therefore $r > 0$. Likewise we have $r < m$. Considering $(mp)(rm)$ gives:

$$\begin{aligned} m^2 rp &= (x_1 y_1 + x_2 y_2 + x_3 y_3 + x_4 y_4)^2 + (x_1 y_2 - x_2 y_1 + x_3 y_4 - x_4 y_3)^2 \\ &\quad + (x_1 y_3 - x_3 y_1 - x_2 y_4 + x_4 y_2)^2 + (x_1 y_4 - x_4 y_1 + x_2 y_3 - x_3 y_2)^2 \end{aligned}$$

Using $x_i \equiv y_i \pmod{m}$, one sees that each term on the right hand side is divisible by m^2 . Dividing both sides by m^2 shows that rp is a sum of four squares and we are done. $\square$

Theorem 8.5. *Every prime p is a sum of four squares.*

Proof. By Corollary 8.3, there exists $0 < m < p$ such that mp is a sum of four squares. By lemma 8.4. If $1 < m$, then we there exists $0 < n_1 < m$ such that n_1p is a sum of four squares. We may continue this process until $n_k = 1$, which gives p as a sum of four squares. Done. $\square$

8.1 Problems

Problem 8.1. Show that the product of sum of four squares is again a sum of four squares, and conclude Lagrange's Four Squares Theorem.

Chapter 9

Generating Functions with applications in Number Theory

Questions about addition naturally lead to the concept of generating functions. The connection is made via the fact that adding m and n are the same as multiplying the two monomials z^m and z^n . For example, we can restate the Lagranges's theorem, which asserts that every positive integer can be written as a sum of four squares as the following:

All the coefficients in

$$f(x) = \left(\sum_{k=0}^{\infty} z^{k^2} \right)^4 = (1 + z + z^4 + z^9 + \dots)^4$$

are positive.

9.1 Change Making and Crazy Dice

Example 9.1. How many ways can we make change for n if the coins are 1, 2 and 3?

Proof. The answer is simply the coefficient of x^n in the expansion:

$$(1+z+z^2+z^3+\dots)(1+z^2+z^4+\dots)(1+z^3+z^6+\dots) = \frac{1}{(1-z)(1-z^2)(1-z^3)} = \sum C(n)z^n,$$

where we assume that $|z| < 1$. Next, we use partial fraction to get:

$$\frac{1}{(1-z)(1-z^2)(1-z^3)} = \frac{1}{6} \frac{1}{(1-z)^3} + \frac{1}{4} \frac{1}{(1-z)^2} + \frac{1}{4} \frac{1}{1-z^2} + \frac{1}{3} \frac{1}{1-z^3}.$$

Thus, since

$$\frac{1}{(1-z)^2} = \frac{d}{dz} \frac{1}{1-z} = \frac{d}{dz} \sum z^n = \sum (n+1)z^n$$

and

$$\frac{1}{(1-z)^3} = \frac{d}{dz} \frac{1}{2(1-z)^2} = \frac{d}{dz} \sum \frac{n+1}{2} z^n = \sum \frac{(n+2)(n+1)}{2} z^n,$$

$$C(n) = \frac{(n+2)(n+1)}{12} + \frac{n+1}{4} + \frac{\chi_1(n)}{4} + \frac{\chi_2(n)}{3},$$

where $\chi_1(n) = 1$ if $2 \mid n$ and $n = 0$ otherwise; $\chi_2(n) = 1$ if $3 \mid n$ and $n = 0$ otherwise. $\square$

Example 9.2. An ordinary pair of dice consists of two cubes each numbered 1 through 6. When tossed together there are altogether 36 (equally likely) outcomes. Thus the sum goes from 2 to 12 with varied numbers of repeats for these possibilities. For example, there are 3 ways for the 10 to show up. Is there any *other* way to number these two cubes with positive integers such that they function just like ordinary pair of dice?

Proof. The question is asking whether there exists $a_1, a_2, \dots, a_6; b_1, b_2, \dots, b_6$, such that:

$$(z^{a_1} + z^{a_2} + \dots + z^{a_6})(z^{b_1} + z^{b_2} + \dots + z^{b_6}) = (z + z^2 + z^3 + z^4 + z^5 + z^6)^2.$$

Let's factorize the right side as $z^2(1+z+z^2)^2(1+z)^2(1-z+z^2)^2$, and called the polynomial consists of powers of a_i 's, and b_i 's the $a(z)$ and $b(z)$ respectively.

Since the a 's and b 's are to be positive and so z must divide both $a(z)$ and $b(z)$. Since $a(1) = b(1) = 6$, $(1+z)(1+z+z^2)$ must divide both $a(z)$ and $b(z)$. All that is left to distribute are the two factors of $1-z+z^2$. If $1-z+z^2$ also divides both $a(z)$ and $b(z)$, then we get the ordinary dice. Therefore, we try to let $a(z) = z(1+z)(1+z+z^2)(1-z+z^2)^2 = z + z^3 + z^4 + z^5 + z^6 + z^8$, and this works! Consequently, we have $b(z) = z + 2z^2 + 2z^3 + z^4$. Translating back, the two dice are marked with 1, 3, 4, 5, 6, 8 and 1, 2, 2, 3, 3, 4 respectively. $\square$

Example 9.3. For which n can there be nonnegative integers $a_1 < a_2 < \dots < a_n$ such that the differences $a_i - a_j, i > j$, take on ALL the values $1, 2, 3, \dots, \binom{n}{2}$. For example, for $n = 4$, the set $\{0, 1, 4, 6\}$ works.

Proof. We claim that $n = 4$ is the only possible n . Let $A(z) = \sum_{i=1}^n z^{a_i}$. In order to get the differences, we multiply $A(z)$ by $A(1/z)$. Thus

$$A(z)A(1/z) = \sum_{i,j=1}^n z^{a_i - a_j},$$

and we split this double sum as $i > j, i = j$, and $i < j$. We obtain:

$$A(z)A(1/z) = \sum_{i,j=1, i>j}^n z^{a_i - a_j} + n + \sum_{i,j=1, i<j}^n z^{a_i - a_j}.$$

To solve our original problem, we need

$$\sum_{i,j=1, i>j}^n z^{a_i-a_j} = \sum_{k=1}^N z^k,$$

where $N = \binom{n}{2}$. Now since the last sum is the same as the first sum, with $1/z$ replacing z , our equation takes the simpler form:

$$A(z)A(1/z) = \sum_{k=-N}^N z^k + n - 1 = \frac{z^{N+1} - z^{-N}}{z - 1} + n - 1.$$

Now we let $z = e^{it}$, (i.e. z lies on the unit circle.) so the left side is simply $|A(z)|^2$, while the right side is:

$$\frac{z^{N+1/2} - z^{-(N+1/2)}}{z^{1/2} - z^{-1/2}} + n + 1 = \frac{\sin(N + 1/2)t}{\sin t/2} + n - 1.$$

Therefore, the equation reduces to:

$$|A(e^{it})|^2 = \frac{\sin \frac{n^2-n+1}{2}t}{\sin t/2} + n - 1.$$

To obtain a contradiction, it suffices to pick t such that

$$\frac{\sin \frac{n^2-n+1}{2}t}{\sin t/2} < -(n - 1).$$

We can choose t such that $\sin \frac{n^2-n+1}{2}t = -1$, for example by picking $t = \frac{3\pi}{n^2-n+1}$. In that case $\sin t/2 < t/2$, $1/\sin(t/2) > 2/t$, $-1/\sin(t/2) < -2/t = -\frac{2n^2-2n+2}{3\pi}$ and so the inequality follows from $-\frac{2n^2-2n+2}{3\pi} < -(n-1)$ or $2n^2-2n+2 > 3\pi(n-1)$. But $2n^2-2n+2-3\pi(n-1) > 2n^2-2n+2-10(n-1) = 2(n-3)^2-6 \geq 2 \times 2^2-6 = 2$, for $n \geq 5$. Hence there is no other n that works.

□

9.2 Representation Function

We introduce the notion of the representation function. Suppose there is a set A of nonnegative integers and that we wish to express the number of ways in which a given integer n can be written as the sum of two of them. The trouble is that we must decide on conventions. Does order count? can the two summands be equal? To remedy this, we define the following:

Definition 9.1.

$$r_A(n) = |\{(a, a') \mid a, a' \in A, n = a + a'\}|.$$

$$r_{A+}(n) = |\{(a, a') \mid a, a' \in A, a \leq a', n = a + a'\}|.$$

$$r_{A-}(n) = |\{(a, a') \mid a, a' \in A, a < a', n = a + a'\}|.$$

In terms of the generating function for the set A , namely $A(z) = \sum_{a \in A} z^a$, we can express the generating functions of these representation functions. We have:

$$\sum r(n)z^n = A^2(z).$$

For $r_{A-}(n)$, we must subtract $A(z^2)$ from $A^2(z)$ to remove the case of $a = a'$ and then divide by 2 to remove the order. Thus, we have:

$$\sum r_{A-}(n)z^n = \frac{1}{2}[A^2(z) - A(z^2)].$$

Finally, for $r_{A+}(n)$, we must add $A(z^2)$ to this result to reinstate the case of $a = a'$, and we obtain:

$$\sum r_{A+}(n)z^n = \frac{1}{2}[A^2(z) + A(z^2)].$$

Example 9.4. Is it possible to design a nontrivial set A , such that $r_{A+}(n)$ is constant?

Proof. The answer is NO. First of all, we must have $0 \in A$, otherwise $r_{A+}(0) = 0$, which means $r_{A+}(n) = 0$ for all n , implying that A is trivial. Now, if $0 \in A$, we must have $1 \in A$, otherwise $r_{A+}(1) \neq r_{A+}(0)$. Then $2 \notin A$, else $r_{A+}(2) = 2$. Then $3 \in A$, else $r_{A+}(3) = 0$, then $4 \notin A$, else $r_{A+}(4) = 2$. Continuing in this manner, we find $5 \in A$. This is a contradiction since now $6 = 1 + 5 = 3 + 3$, and $r_{A+}(6) = 2$. $\square$

Example 9.5. Is it possible to find a nontrivial infinite set A , such that $r_{A+}(n)$ is eventually constant?

Proof. Using the generation of $r_{A+}(n)$, the problem reduces to whether there is an infinite set A for which:

$$\frac{1}{2}[A^2(z) + A(z^2)] = P(z) + \frac{C}{1-z},$$

where $P(z)$ is a polynomial.

Now taking limit as $z \rightarrow (-1)^+$. Clearly $P(z)$ and $C/(1-z)$ remain bounded, $A^2(z)$ remains nonnegative, and $A(z^2)$ goes to $A(1) = \infty$, a contradiction. $\square$

Example 9.6. Can we split the nonnegative integers in two sets A and B such that every integer n is expressible in the same numbers of ways as the sum of two distinct members of A , as it is as the sum of two distinct members of B ?

Proof. Basically, we are asking whether there exists $A, B \subset \mathbb{Z}_{\geq 0}$ that partitions $\mathbb{Z}_{\geq 0}$ such that $r_A(n) = r_B(n)$ for all n .

If we turn to generating functions, we are solving the following simultaneous functional equations:

$$A(z) + B(z) = \frac{1}{1-z}.$$

$$\frac{1}{2}[A^2(z) - A(z^2)] = \frac{1}{2}[B^2(z) - B(z^2)].$$

From this, we have:

$$[A(z) - B(z)]\frac{1}{1-z} = A(z^2) - B(z^2),$$

or

$$A(z) - B(z) = (1-z)[A(z^2) - B(z^2)].$$

We now iterate the process, and we get:

$$A(z^2) - B(z^2) = (1-z^2)[A(z^4) - B(z^4)],$$

so that continuing gives:

$$A(z) - B(z) = (1-z)(1-z^2) \dots (1-z^{2^{n-1}})[A(z^{2^n}) - B(z^{2^n})].$$

Now, letting $n \rightarrow \infty$, and using $A(0) = 1$, and $B(0) = 0$, we have:

$$A(z) - B(z) = \prod_{i=0}^{\infty} (1 - z^{2^i}).$$

This product is easy to multiply out. Every term z^n occurs uniquely since every n is uniquely the sum of distinct powers of 2. Indeed, z^n occurs with coefficient $+1$ if n is the sum of an even number of distinct powers of 2, and it has coefficient -1 , otherwise.

Therefore, the sets A, B exists, and are unique. Indeed, A consists of nonnegative integers which are the sum of an even number of powers of 2, and B consists of nonnegative integers which are the sum of an odd number of distinct powers of 2.

□

Example 9.7. Let a, b , and c be integers satisfying $0 < a < c - 1$ and $1 < b < c$. For each k , $0 \leq k \leq a$, let $r_k, 0 \leq r_k < c$ be the remainder of kb when divided by c . Prove that the two sets $\{r_0, r_1, r_2, \dots, r_a\}$ and $\{0, 1, 2, \dots, a\}$ are different.

Proof. Let $B(x) = 1 + x^b + x^{2b} + \dots + x^{ab}$ and $A(x) = 1 + x + x^2 + \dots + x^a$. Suppose the two sets are equal, we first show that $(b, c) = 1$. Let $d = (b, c)$, then $d \mid r_k$ for all k . Hence $d \mid 1$, we have $d = 1$ as claimed. Let $f(x) = A(x) - B(x)$. Writing $kb = q_k c + r_k$ gives: $x^{kb} - x^{r_k} = x^{q_k c + r_k} - x^{r_k} = x^{r_k}(x^{q_k c} - 1)$ and since $x^c - 1$ divides $x^{q_k c} - 1$, we have $x^c - 1$ divides $f(x)$.

From

$$f(x) = \frac{x^{(a+1)b} - 1}{x^b - 1} - \frac{x^{a+1} - 1}{x - 1} = \frac{F(x)}{(x - 1)(x^b - 1)},$$

where $F(x) = x^{ab+b+1} + x^b + x^{a+1} - x^{ab+b} - x^{a+b+1} - x$, we have:

$$F(x) \equiv 0 \pmod{x^c - 1}.$$

Since $x^c \equiv 1 \pmod{x^c - 1}$, we conclude that

$$\{ab + b + 1, b, a + 1\} \equiv \{ab + b, a + b + 1, 1\} \pmod{c}.$$

Thus $b \equiv ab + b, a + b + 1$ or $1 \pmod{c}$.

However, since $1 < b < c$ and $0 < a < c - 1$, neither $b \equiv 1 \pmod{c}$ nor $b \equiv a + b + 1 \pmod{c}$ are possible. Therefore, we must have $b \equiv ab + b \pmod{c}$. Since $(b, c) = 1$, we have $1 \equiv a + 1 \pmod{c}$, which is again impossible since $0 < a < c - 1$ and we have a contradiction. Therefore, the two sets are not equal. $\square$

9.3 Problems

Problem 9.1. Write down the generating function representing the following:

- The number of partitions of n into parts which are odd.
- The number of partitions of n into parts which are even.
- The number of partitions of n into parts which are squares.
- The number of partitions of n into parts which are primes.
- The number of partitions of n into parts which are unequal.
- The number of partitions of n into parts which are odd and unequal.
- The number of partitions of n into parts which are even and unequal.

(h) The number of partitions of n into parts which are distinct squares.

(i) The number of partitions of n into parts which are distinct primes.

Problem 9.2. Prove the theorem of Euler: The number of ways of expressing n as the sums of distinct positive integers equals the number of ways of expressing n as the sum of (not necessarily distinct) odd positive integers.

Problem 9.3. (a) Produce a set A such that $r(n) > 0$ for all n in $1 \leq n \leq N$, but with $|A| \leq \sqrt{4N+1}$.

(b) Show that every set satisfying the conditions of (a) must have $|A| \geq \sqrt{N}$.

Problem 9.4. Can the positive integers be split into at least two arithmetic progressions such that any two of which have a distinct common difference?

Chapter 10

Arithmetic Functions and Average Orders

Chapter 11

Continued Fractions

Chapter 12

Quadratic Forms

Chapter 13

On ABC Conjecture

13.1 ABC Conjecture

Notation 13.1. $\prod_{p|N} p$ denotes the product of primes dividing N

Conjecture 13.2 (ABC Conjecture). *Suppose $A, B, C \in \mathbb{N}$, such that $(A, B, C) = 1$ and $A + B = C$. Then given $\epsilon > 0$, there exists a positive constant $K(\epsilon)$ such that:*

$$C \leq K(\epsilon) \left(\prod_{p|ABC} p \right)^{1+\epsilon}.$$

Conjecture 13.3 (Variant of ABC Conjecture). *With the same assumption as ABC Conjecture, we have:*

$$\max\{|A|, |B|, |C|\} \leq K(\epsilon) \left(\prod_{p|ABC} p \right)^{1+\epsilon}.$$

Conjecture 13.4 (Weak ABC Conjecture). *With the same assumption as ABC conjecture, we have:*

$$|ABC|^{1/3} \leq K(\epsilon) \left(\prod_{p|ABC} p \right)^{1+\epsilon}.$$

Example 13.1. ABC conjecture implies there that exists an integer N such that Fermat's equation has no non-trivial solution for $n > N$.

Proof. We may WOLOG assume $(x, y, z) = 1$. If $n \geq 2$, then $z \geq 3$. Choose $\epsilon = 1$, let $K = \max\{1, K(1)\}$, then ABC conjecture implies:

$$z^n \leq K \left(\prod_{p|xyz} \right)^2 < K z^6.$$

Thus we have:

$$n < 6 + \frac{\log K}{\log z} \leq 6 + \frac{\log K}{\log 3}.$$

We are done. □

Example 13.2. ABC conjecture implies Catalan's equation has only finitely many solutions.

Proof. Let (a, b, c, d) be a solution to the Catalan equation with $\min b, d \geq 3$. Then $(a, b) = 1$. Let $\epsilon = 1/4$. Then by ABC conjecture, we have:

$$c^d < a^b \leq K(1/4) \left(\prod_{p|ac} \right)^{5/4} \leq K(1/4)(ac)^{5/4}.$$

This implies:

$$\begin{aligned} b \log a &\leq \log K(1/4) + \frac{5}{4}(\log a + \log c) \\ d \log c &< \log K(1/4) + \frac{5}{4}(\log a + \log c). \end{aligned}$$

Combining the two inequalities and rearrange gives:

$$\left(b - \frac{5}{2}\right) \log a + \left(c - \frac{5}{2}\right) \log d < 2 \log K(1/4).$$

Since $a, b \geq 2$. We have the inequality:

$$c + d < 2 \log K(1/4) / \log 2 + 5.$$

Thus there are only finitely many pairs of exponents for which the Catalan equation is solvable. Finally, for fixed exponents $b, d \geq 3$. There are only finitely many solutions to the Catalan's equation and we are done. [This last sentence is true since ABC implies $a^b \leq K(1)(ac)^2 = K(1)a^2(a^b - 1)^{2/d} \leq K(1)a^2(a^b - 1)^{2/3}$. This inequality only holds for finitely many a 's are we are done.] □

Example 13.3. ABC conjecture is false if we omit the condition $(A, B, C) = 1$.

Proof. Let $A = 3^n, B = 23^n, C = 3^{n+1}$. Suppose ABC conjecture still holds. Then for $\epsilon = 1$, there exists constant $K(1)$ such that $3^{n+1} \leq K(1)6^2$, for all n . This is clearly absurd for sufficiently large n . $\square$

13.2 ABC Conjecture for Function Fields

Theorem 13.5 (Mason). *Let $a, b, c \in \mathbb{C}[t]$, not all constant such that $(a, b, c) = 1$ and $a + b = c$. Then we have:*

$$\max\{\deg a, \deg b, \deg c\} \leq n_0(abc) - 1,$$

where $n_0(f)$ denotes the number of distinct roots of f .

Proof. Write $a = a_m \prod_{i=1}^{n_0(a)} (t - \alpha_i)^{m_i}, b = b_n \prod_{i=1}^{n_0(b)} (t - \beta_i)^{n_i}, c = c_r \prod_{i=1}^{n_0(c)} (t - \gamma_i)^{r_i}$. Let $u = a/c, v = b/c \in \mathbb{C}(t)$, and define $L(f(t)) = \frac{d}{dt}(\log f(t)) = f'/f$ to be the formal logarithmic derivative. Note that we have $u + v = 1$ and thus we have $uL(u) + vL(v) = u' + v' = (u + v)' = 0$. Now, $L(v) \neq 0$ implies

$$\frac{b}{a} = \frac{v}{u} = -\frac{L(u)}{L(v)}.$$

Logarithmically differentiating the product form of u, v yields:

$$L(u) = L(a/c) = \sum_{i=0}^{n_0(a)} \frac{m_i}{t - \alpha_i} - \sum_{i=0}^{n_0(c)} \frac{r_i}{t - \gamma_i},$$

and

$$L(v) = L(b/c) = \sum_{i=0}^{n_0(b)} \frac{n_i}{t - \beta_i} - \sum_{i=0}^{n_0(c)} \frac{r_i}{t - \gamma_i}.$$

Let $f = \prod_{i=1}^{n_0(a)} (t - \alpha_i) \prod_{i=1}^{n_0(b)} (t - \beta_i) \prod_{i=1}^{n_0(c)} (t - \gamma_i) =$ product of distinct roots of abc , thus $\deg f = n_0(abc)$ since $(a, b, c) = 1$. This implies $fL(u), fL(v)$ are polynomials of degree at most $n_0(abc) - 1$.

We have $b/a = -L(u)/L(v)$ implies $a(fL(u)) = -b(fL(v))$. Since $(a, b) = 1$, we have $a \mid fL(v)$ and hence $\deg a \leq \deg fL(v) \leq n_0(abc) - 1$. Similarly, the same inequality holds for b and c and the assertion follows. $\square$

Theorem 13.6 (FLT for Polynomials). *If $n \geq 3$, then the equation $a^n + b^n = c^n$ has no solution in nonzero, not all constant polynomials $a, b, c \in \mathbb{C}[t]$, and $(a, b, c) = 1$.*

Proof. Let $n \geq 3$, we have $n_0(a^n b^n c^n) = n_0(abc)$. Mason's theorem yields:

$$\deg a^n \leq n \max(\deg a, \deg b, \deg c) \leq n_0(abc) - 1 = \deg a + \deg b + \deg c - 1.$$

It follows that:

$$n(\deg a + \deg b + \deg c) \leq 3(\deg a + \deg b + \deg c) - 3 \leq n(\deg a + \deg b + \deg c) - 3,$$

which is a contradiction. $\square$

13.3 Problems

Problem 13.1. Prove that the two versions of ABC conjectures are equivalent.

Problem 13.2. Prove that ABC conjecture implies Weak ABC conjecture.

Problem 13.3. Prove that the equation $x^m - y^n = 1$ has no solution in non-constant polynomials $x, y \in \mathbb{C}[t]$ and integers $m, n \geq 2$.

Chapter 14

A Brief Introduction to Elliptic Curves and Modular Forms

Over the past decades, theory of elliptic curves over finite fields has been applied to various problems in cryptography. For example, factoring large integers, primality testing, and cryptosystem. In this section, we shall give a brief survey of basic theory of elliptic curves over the complex numbers, the rational numbers, and finite fields.

14.1 Elliptic Curves

An elliptic curve over a field $\mathbb{F}$ is curve given by the equation:

$E : y^2 + a_1xy + a_3 = x^3 + a_2x^2 + a_4x + a_6$, where $a_i \in \mathbb{F}$ satisfying one extra condition that the curve non-singular. That is to say that the partial derivatives cannot both vanishes at the same point. This means:

$$a_1y = 3x^2 + 2a_2 + a_4, \quad 2y + a_1x = 0$$

cannot be solved simultaneously over the algebraic closure of $\mathbb{F}$.

In the case that $\text{char}(\mathbb{F}) \neq 2, 3$, we may make a change of variables and use the representation:

$$E : y^2 = x^3 + ax + b, a, b \in \mathbb{F}$$

instead. The condition that the curve is non-singular is equivalent to the cubic on the right having three distinct roots. From now on, we shall assume that $\text{char}(\mathbb{F}) \neq 2, 3$.

We let $E(\mathbb{F}) = \{(x, y) \in \mathbb{F} \times \mathbb{F} \mid y^2 = x^3 + ax + b\} \cup \{O\}$, here the point $\{O\}$ is the so-called “point at infinity”.

14.1.1 The Group Law

One of the reasons why people study elliptic curves is that they are equipped with a group law, that is defined as follows:

Let the curve be $E : y^2 = f(x)$, where $f(x) = x^3 + ax + b, a, b \in \mathbb{F}$. Let $P = (x_1, y_1), Q = (x_2, y_2) \in E(\mathbb{F})$. Connect the two points by a straight line. Since $f(x)$ is a cubic, this line will cross at a third point, and we call it $P * Q$. Then we reflect the point $P * Q$ about the x -axis, and call this new point on the curve $P + Q$.

Remark: The reflection is really connecting the point $P * Q$ to the point O at infinity, and the point $P + Q$ is the third intersection point.

Now, one may check that under this definition, $E(F)$ is an abelian group with the identity element being the point at infinity O . We can briefly describe why it is a group. Checking closure of addition is a simple, but tedious exercise in high school algebra. For inverses, one can check easily that if $P = (x, y)$, then $-P = (x, -y)$. Commutativity is obvious, and associativity is the hardest to check. One can either brute force using high school algebra, or there is a nice proof using Picard group, but we will not discuss this here.

Remark To double a point P , we simply take the tangent line at P , the tangent will again cross the curve again at a point Q , then reflection of Q about the x -axis is the point $2P$.

14.1.2 Elliptic Curves over $\mathbb{C}$

Let $E : y^2 = 4x^3 - g_2x - g_3, a, b \in \mathbb{C}$. We now study the structure of the group $E(\mathbb{C})$. To this end, we have the following theorem:

Theorem 14.1. *Let E be an elliptic curve defined over $\mathbb{C}$. Then we have:*

$$\mathbb{C}/L \cong E(\mathbb{C}),$$

where $L = \{aw_1 + bw_2 \mid a, b \in \mathbb{Z}, w_1, w_2 \text{ } \mathbb{R}\text{-linearly independent}\}$

This isomorphism is given by the map: $z + L \mapsto (\wp(z), \wp'(z))$, where

$$\wp(z) = \frac{1}{z^2} + \sum_{\ell \in L, \ell \neq 0} \frac{1}{(z - \ell)^2} - \frac{1}{\ell^2}$$

is the Weierstrass $\wp$ -function, which is meromorphic in $\mathbb{C}$ with double poles precisely at the points in L .

Remark: $g_2 = g_2(L) = 60 \sum_{\ell \in L} 1/\ell^4$, $g_3 = g_3(L) = 140 \sum_{\ell \in L} 1/\ell^6$.

Remark: Topologically, $\mathbb{C}/L$ is a torus. This can be viewed as a square with the opposites being identified.

14.1.3 Elliptic Curves over $\mathbb{Q}$

We now turn our gear to study elliptic curves over $\mathbb{Q}$. We should note that studying the arithmetic of elliptic curves remains one of the most active areas of modern number theory. While, the complex points of an elliptic curve forms a torus, the situation is quite different in the rational case. We have the following theorem of Mordell-Weil:

Theorem 14.2 (Mordell-Weil). *Let E be an elliptic curve over $\mathbb{Q}$, then $E(\mathbb{Q})$ is a finitely generated abelian group. In other words,*

$$E(\mathbb{Q}) \cong \mathbb{Z}^r \times T,$$

where T is a finite group.

Therefore, the group $E(\mathbb{Q})$ breaks up into two parts. The integer r is the so-called rank of the elliptic curve E , and T is referred as the torsion subgroup of $E(\mathbb{Q})$.

Of the two parts, the structure of the torsion subgroup is completely determined. The following is a deep theorem by Mazur that characterizes the torsion subgroup:

Theorem 14.3 (Mazur). *Let E be an elliptic curve over $\mathbb{Q}$, then the torsion subgroup T of $E(\mathbb{Q})$ is one of the following groups:*

$$\begin{aligned} \mathbb{Z}/N\mathbb{Z} \quad 1 \leq N \leq 10 \quad \text{or} \quad N = 12 \\ \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2N\mathbb{Z} \quad 1 \leq N \leq 4 \end{aligned}$$

While for the rank r , this remains to be one of the most mysterious objects in modern number theory. It is usually very difficult to determine the rank of an elliptic curve precisely. What people usually do is to study a rather a group called Selmer group, which is closely related to $E(\mathbb{Q})$, that is a little easier to work with.

14.1.4 Elliptic Curves over $\mathbb{F}_q$

Finally, we move on to case where $\mathbb{F} = \mathbb{F}_q$ is a finite field with q elements. In this case, the group $E(\mathbb{F}_q)$ is in some sense considerably easier, since it's simply:

$$E(\mathbb{F}_q) \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$$

for some integers $m, n \geq 1$ with $(m, q) = 1$.

Now, we restrict ourselves to the case where $q = p$ a prime. Consider $E : y^2 = x^3 + ax + b, a, b \in \mathbb{F}_p$. It is clear that we can compute the order of $E(\mathbb{F}_p)$ by simply counting. Let $a_p = p + 1 - |E(\mathbb{F}_p)|$, then we have the following Hasse bound:

Theorem 14.4 (Hasse). *Let E be an elliptic curve over $\mathbb{F}_q$, then we have:*

$$|a_q| \leq 2\sqrt{q}.$$

This theorem is related to the so-called Riemann Hypothesis for elliptic curves, but we will not discuss this here.

We now go back to elliptic curves over $\mathbb{Q}$ and define the Hasse-Weil L -series for elliptic curves.

For an elliptic curve $E : y^2 = x^3 + ax + b$ over $\mathbb{Q}$. We reduce each of the coefficient modulo a prime p and call the reduced curve $\tilde{E} : y^2 = x^3 + \bar{a}x + \bar{b}$, where $\bar{a}, \bar{b}$ are the image of a, b of the reduction mod p map. Then, we compute $|\tilde{E}(\mathbb{F}_p)|$, and let $a_p = p + 1 - |\tilde{E}(\mathbb{F}_p)|$. Let $\Delta = -(4a^3 + 27b^2)$, be the discriminant of the elliptic curve E .

Definition 14.5 (Incomplete L-Series). *The incomplete L-series of E is*

$$L(E, s) = \prod_{p \nmid \Delta} \frac{1}{1 - a_p p^{-s} + p^{1-2s}},$$

where the product runs over all primes not dividing Δ .

It was conjectured that this incomplete L -series has analytic continuation to the whole complex plane. This conjecture is a consequence of the Taniyama-Shimura conjecture, which asserts that every elliptic curve is modular. We will talk more about this in the final section.

14.2 Elliptic Curve Encryption

Elliptic Curve Cryptosystems were invented independently by Victor Miller and Neal Koblitz in 1985. In this section, we describe how the theory of elliptic curves over finite can be applied to obtain cryptosystem. We begin with the classical Diffie-Hellman Key Exchange.

14.2.1 Classical Diffie-Hellman

Definition 14.6 (Discrete Log Problem). *The discrete log problem in the group G to the base $g \in G$ is the problem, given $y \in G$, of finding an integer x such that $g^x = y$ ($xg = y$ when the group operation in G is additive), provided that such an integer exists (in other words, provided that $y \in \langle g \rangle$).*

In the case of classical Diffie-Hellman, we will let $G = \mathbb{F}_p^\times$, the multiplicative group of a finite field, which is known to be cyclic.

The Diffie-Hellman key exchange works as follows. Suppose that Alice and Bob want to agree upon a large integer to serve as a key for some private key cryptosystem. This must be done using open communication channels - that is, any eavesdropper (Eve) knows everything that Alice sends to Bob and everything that Bob sends to Alice. Alice and Bob first agree on a prime p and a base element $g \in \mathbb{F}_p^\times$. This has been agreed upon publicly, so that Eve also has this information at her disposal. Next, Alice secretly chooses a random positive integer $k_A < p$ (of about the same magnitude as p), computes the least positive residue mod p of g^{k_A} (by repeated squaring method), and sends this to Bob. Meanwhile, Bob does likewise: he sends $g^{k_B} \in \mathbb{F}_p^\times$ to Alice, while keeping k_B secret. The agreed upon key will then be the integer

$$g^{k_A k_B} \in \mathbb{F}_p^\times$$

which Bob can compute by raising the integer he received from Alice to his secret k_B -power in $\mathbb{F}_p$, and Alice can compute by raising the integer she received from Bob to the k_A -power in $\mathbb{F}_p$. This works because in $\mathbb{F}_p^\times$ we have

$$g^{k_A k_B} = (g^{k_A})^{k_B} = (g^{k_B})^{k_A}.$$

The problem that Eve is facing is the so-called Diffie-Hellman problem: Given $g, g^{k_A}, g^{k_B} \in \mathbb{F}_p^\times$, find $g^{k_A k_B}$.

Remark: It is easy to see that anyone who can solve the discrete log problem in $\mathbb{F}_p^\times$ can then immediately solve the Diffie-Hellman problem as well. The converse is not known. That is, it is conceivable (though thought to be unlikely) that someone could invent a way to solve the Diffie-Hellman problem without being able to find discrete logarithms. In other words, breaking the Diffie-Hellman key exchange has not been proved to be equiv- equivalent to solving the discrete log problem. It is generally believed that for practical purposes it is probably safe to assume that the Diffie-Hellman key exchange is secure provided that the discrete logarithm problem is intractable.

14.2.2 Elliptic Curve Diffie-Hellman

The classical Diffie-Hellman key exchange can be adapted for elliptic curves as follows. First note that a "random" point on an elliptic curve E can serve as a key, since Alice and Bob can agree in advance on a method to convert it to an integer (for example, they can take the image of its x -coordinate under some agreed upon simple map from $\mathbb{F}_p$ to the natural numbers). So suppose that E is an elliptic curve over $\mathbb{F}_p$, and Q is an agreed upon (and

publicly known) point on the curve. Alice secretly chooses a random integer k_A , **the secret key of Alice**, and computes the point $k_A Q$, which she sends to Bob. Likewise, Bob secretly chooses a random k_B , **the secret key of Bob**, and computes $k_B Q$, and sends it to Alice. The **common key** is

$$P = k_A k_B Q.$$

Alice computes P by multiplying the point she received from Bob by her secret k_A . Bob computes P by multiplying the point he received from Alice by his secret k_B . An eavesdropper who wanted to spy on Alice and Bob would have to determine $P = k_A k_B Q$ knowing Q , $k_A Q$, and $k_B Q$, but not k_A or k_B . The eavesdropper's task is called the "Diffie-Hellman problem for elliptic curves".

In the case $G = E(\mathbb{F}_p)$, the elliptic curve discrete log problem to the base $Q \in E(\mathbb{F}_p)$ is the problem, given $P \in E(\mathbb{F}_p)$, of finding an integer x such that $P = xQ$ if such x exists.

Remark: Just like the classical case, it is easy to see that the elliptic curve Diffie-Hellman problem can be solved if the elliptic curve discrete log problem can be solved. Namely, the eavesdropper, who knows Q and $k_A Q$, finds the secret k_A and then has broken the cipher. The converse - the assertion that the elliptic curve Diffie-Hellman problem is equivalent to the elliptic curve discrete log problem - is a conjecture but has not been proved.

Remark: No subexponential time algorithm is known in general for the case of elliptic curves Diffie-Hellman problem. The only methods available for finding discrete logs on $E/\mathbb{F}_p$ are the methods that apply to arbitrary groups. All of them have running time of the form $L_p(l) = p^{O(1)} = \exp(O(\ln p))$, provided that $|E(\mathbb{F}_p)|$ is divisible by a large prime ("large" means that its order of magnitude is not much less than that of p).

14.3 Problems Related Elliptic Curves

14.3.1 Fermat's Last Theorem

Theorem 14.7 (FLT). *Let $n \geq 3$ be an integer. Then the equation $x^n + y^n = z^n$ has no positive integer solutions.*

The theorem is first conjectured by Fermat in 1637, and after the collaborations of numerous mathematicians, it is finally solved by Andrew Wiles in 1994. His proof uses a lot of advanced techniques in modern number theory, including theories of elliptic curves. We shall give a brief sketch of his proof.

First of all, we note that it suffices to prove FLT for a prime exponent p . Now suppose (a, b, c) is a solution to the Fermat's equation. In 1985, Gerhart Frey considered the elliptic curve $E : y^2 = x(x - a^p)(x + b^p)$. He then observed that the existence of this curve will contradict the Taniyama-Shimura conjecture. In the same year, Jean-Pierre Serre, and Ken Ribet showed that Taniyama-Shimura conjecture implies Fermat's Last Theorem. Before getting more into this, we shall briefly describe the Taniyama Shimura conjecture:

Theorem 14.8 (Taniyama-Shimura Conjecture). *(Proved by Breuil, Conrad, Diamond, Taylor, and Wiles) Every elliptic curve over $\mathbb{Q}$ is modular.*

What does an elliptic curve being modular means? It means that following: Start with an elliptic curve E over $\mathbb{Q}$. We then form it's complete L -series, $L(E, s)$. (The difference between the incomplete L -series and the complete L -series only off a some constant multiple, and we shall not take the trouble to define the complete L -series.) We write:

$$L(E, s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s},$$

after expanding the infinite product.

We now form a complex function:

$$f_E(q) = \sum_{n=1}^{\infty} a_n q^n,$$

where $q = e^{2\pi iz}$, with z lies in the upper half plane in $\mathbb{C}$. The elliptic curve E being modular means that this function $f_E(q)$ is a modular form. To explain what is a modular form, we first introduce some special subgroups of $\text{SL}_2(\mathbb{Z})$:

Definition 14.9.

$$\Gamma_0(N) = \left\{ \gamma \in \text{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{N} \right\}$$

We then define weakly modular functions :

Definition 14.10. *Let $k \in \mathbb{Z}$. A meromorphic function $f : \mathbb{H} \rightarrow \mathbb{C}$ is **weakly modular of weight k with respect to Γ** if:*

$$f(\gamma(t)) = (ct + d)^k f(t), \text{ for } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma,$$

where $\mathbb{H}$ denote the complex upper half plane. i.e. $\mathbb{H} = \{z \in \mathbb{C} | \text{im } z > 0.\}$

Note: Γ can be taken to be $\Gamma_0(N)$ for some N .

Definition 14.11. Let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a weakly modular function of weight k with respect of Γ , then f is a **modular form of weight k with respect of Γ** if:

$$f(q) = \sum_{n=0}^{\infty} a_n(f) q^n, q = e^{2\pi iz}.$$

Basically, Andrew Wiles proved the Fermat's Last Theorem by proving an important special case of Taniyama-Shimura conjecture, which is enough to conclude the Fermat's Last Theorem.

14.3.2 Birch and Swinnerton-Dyer Conjecture

This is one of the seven millenium problems by the Clay Mathematical Institute. We conclude the survey by stating the conjecture:

Let E be an elliptic curve over $\mathbb{Q}$. By Mordell-Weil theorem, we have $E(\mathbb{Q}) \cong \mathbb{Z}^r \times T$. On the other hand, we look at the incomplete L -series $L(E, s)$ for E . Let r^{an} denotes the order of vanishing of $L(E, s)$ at 1. In other words: $r^{an} = ord_{s=1}(E, s)$.

Theorem 14.12 (Birch and Swinnerton-Dyer Conjecture). *Let E be an elliptic over $\mathbb{Q}$, then we have:*

$$\dim_{\mathbb{Q}}(E(\mathbb{Q}) \otimes_{\mathbb{Z}} \mathbb{Q}) = r = r^{an} = ord_{s=1} L(E, s).$$

Chapter 15

The Modularity Theorem and the Fermat's Last Theorem

We end this book by a briefly explaining the statement of the modularity theorem, also known as the Taniyama-Shimura Conjecture, and how this help proving the Fermat's Last Theorem.

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